Master_Notes

Electronics Notes

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Random Variable

- Random Experiment:-

A setup where you know all possible outcomes but can't predict the exact outcome before it happens.

Eg: (a) Tossing a fair coin. → {H, T}

(b) Rolling a fair dice. → {1, 2, 3, 4, 5, 6}

(c) Tossing a coin which is both sided "head." {H}

- Sample space:-

List of all possible outcomes of a random experiment.

Eg: (a) Tossing a fair coin.

$$S = \{H, T\}$$

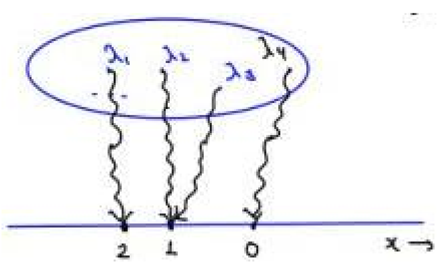
(b) Rolling a dice.

$$S = \{1, 2, 3, 4, 5, 6\}$$

(c) Tossing two coins simulatenously.

$$S = \{HH, HT, TH, TT\}$$

- Sample point:- Each element of a sample space.
- Random Variable:-



A random variable is a function that assigns a real number to each sample point.

Eg: x is a random variable that represent the no. of heads appearing when two coin are tossed simulatenously.

x: no. of heads appearing when two coin tossed simulatenously.

$$S = \{HH, HT, TH, TT\}$$

$$x(HH) = 2$$

$$x(TH) = 1$$

$$x \in \{0, 1, 2\}$$

$$x(HT) = 1$$

$$x(TT) = 0$$

$$S = \{\lambda_1, \lambda_2, \lambda_3 - \dots\}$$

Sample points

$$OR \to +$$

$$x \in \{0, 1, 2\} \to 0.R.V.$$

$$x \in \mathbb{Z}$$

$$P(x - 1) = 0$$

Q. find

$$P(x = 0) = P(TT) = 1/4$$

$$P(x = 1) = P(HT) + P(TH) = \frac{1}{4} + \frac{1}{4} = 1/2$$

$$P(x = 2) = P(HH) = 1/4$$

$$P(x = 3) = 0$$

$$P(|x| \leq 1) = P(-1 \leq x \leq 1)$$

$$= P(x = 0) + P(x = 1)$$

$$= \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

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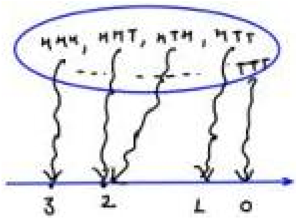
(b) Y is a random variable that represents the no. of heads appearing when three coin are tossed simultaneously.

$$s = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Y: no. of heads appearing



$$Y(HHH) = 3$$

$$Y(HHT) = 2 = Y(HTH) = Y(THH)$$

$$Y(HTT) = 1 = Y(HTT) = Y(TTH)$$

$$Y(TTT) = 0$$

$$Y \in \{0,1,2,3\}$$

$$Y \in \{0,1,2,3\} \rightarrow 0 \text{ R.V.}$$

$$\begin{aligned} & P(TT, HTT, THT, TTH, HHT, HTH, THH \\ Q \text{ Find; } P(Y \leq 3) = & P(Y=0) + P(Y=1) + P(Y=2) + P(Y=3) = 1 \\ & = \tfrac{1}{8} + \tfrac{3}{8} + \tfrac{3}{8} + \tfrac{1}{8} = 1 \end{aligned}$$

$P(Y^2 + 2 \leq 6) = P(-2 \leq Y \leq 2) = P(Y=0) + P(Y=1) + P(Y=2)$
$Y^2 + 2 \leq 6$
$Y^2 \leq 4$
$-2 \leq Y \leq 2$

$\frac{X^2+4}{2} > 10 \qquad X^2 > 16$
$X^2 + 4 > 20 \qquad X > 4; X <$

Q. You roll a fair dice. $\rightarrow s = \{1,2,3,4,5,6\}$ $P(X=10) = P(\text{no. of dice is } 4)$

Random variable x: (no. appearing on the dice)² - 6

Find $P\left[\left(\frac{X^2+4}{2} - 10\right) > 0\right]$
$= P(X > 4) + P(X < -4)$
$= P(X=10) + P(X=13) + P(X=30) + P(X=-5) = 1/6 + 1/6 + 1/6 + 1/6 = 4/6 = \boxed{2/3}$

Q. A group of students typically take 30 min. to 60 min. to solve a Quiz. A random variable x is defined such that it represents the time taken by a randomly selected student to solve the Quiz.

(a) what all values (x) can take? (x \in [30, 60] \rightarrow C \cdot R \cdot V)
(b) (P(X = 35) = \frac{1}{\infty} = 0)

$$P(X = 43.5) = \frac{1}{\infty} = 0$$

Random Variable

Discrete Random Variable (DRV)

$$x \in \{-2, 3, 4.5, 6\}$$

$x \in \{-6, -4.3, 2, 4, 7, 11 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

Mixed R.V.

$$x \in \{1, 2\} + [3, 6]$$

$$x \in \{-3, 4\} + [6, 70]$$

Continuous Random variable

$$x \in [-20, 40]$$

$$x \in [-2.5, 3.5]$$

$$x \in (3, 3.2)$$

N.B. - The probability at a point is zero, in case of CRV

$$P(X = a) = 0$$

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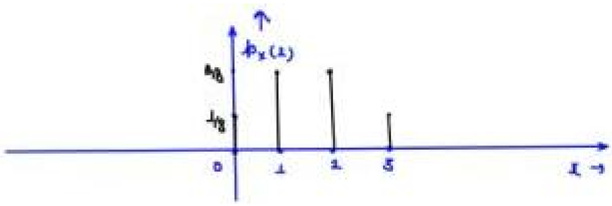
- Probability Mass Function (PMF):

Defines the probability at a point.

PMF:

$$p_x(x) = P(x = x)$$



$$p_x(10) = P(x = 10)$$

Q. Tossing 3 coins simultaneously.

x: no of heads appearing

Draw P.M.F. of R.V. x.

$$x \in \{0, 1, 2, 3\} \rightarrow DRV$$

PMF:

$$P(x = 0) = 1/8$$

;

$$P(x = 1) = \frac{x}{8}$$

,

$$P(x=2) = 3/8$$

,

$$P(x=3) = 1/8$$

x=x	0	1	2	3
p_{x}(x)	1/8	3/8	5/8	1/8

for CRV,

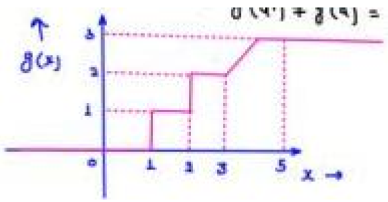
$$P(x=a) = 0$$

for CRV, PMF:

$$p_x(x) = 0$$

- Phasic Mathematics Problem:

$$\forall g(a^+) = g(a) = g(a^-) \rightarrow \text{cont. at } x = a$$
$$\forall g(a^+) = g(a) \neq g(a^-) \rightarrow \text{cont. from R.H.s}$$
$$\forall g(a^+) \neq g(a) = g(a^-) \rightarrow \text{L.H.s}$$



Discuss the continuity

at

$$x = 1, x = 2, x = 3, x = 5$$

Given that

$$g(1) = 1, g(2) = 2$$
$$x = a; \quad g(a^+); \quad g(a), \quad g(a^-)$$
$$g(1) = 1; \quad g(1^+) = 1; \quad g(1^-) = 0; \quad \underbrace{g(1) = g(1^+)}_{\text{NOT cont. at } x=1}$$
$$g(2) = 2; \quad g(2^+) = 2; \quad g(2^-) = 1; \quad \text{NOT cont. at } x = 2$$
$$g(3) = 2; \quad g(3^+) = 2; \quad g(3^-) = 2; \quad \text{cont. at } x = 3 \quad g(4) = g(1^-)$$
$$g(5) = 3; \quad g(5^+) = 3; \quad g(5^-) = 3; \quad \text{cont. at } x = 5$$

- Cumulative Distribution function (CDF):

$$F_x(x) = P[x \leq x]$$

$$\text{E.g. } F_x(2) = P[x \leq 2] \quad F_x(10^2) = P[x \leq 10^2] = 1$$

$$\text{or } F_x(4) = P[x \leq 4] = 1$$

Q. x: no of heads appearing when 3 coins tossed simultaneously.

Draw COF of R.V. x?

$$F_x(0.5) = P[x \leq 0.5] = P[x = 0] = 1/8$$

$$x \in \{0, 1, 2, 3\} \rightarrow DRV$$

$$F_x(1) = P[x \leq 1] = P[x = 0] + P[x = 1] = 1/2$$

$$F_x(-1) = P[x \leq -1] = 0 \quad F_x(1.5) = P[x \leq 1.5] = P[x = 0] + P[x = 1] = 1/2$$

$$F_x(-0.5) = P[x \leq -0.5] = 0 \quad F_x(1) = P[x \leq 2] = P[x = 0] + P[x = 1] + P[x = 2]$$

$$F_x(0.5) = P[x \leq 0.5] = 3/8$$

$$\Rightarrow F_x(0) = P[x \leq 0] = P[x = 0] = 1/8 \quad F_x(3) = P[x \leq 3] = P[x = 0] + P[x = 1] + P[x = 2] + P[x$$

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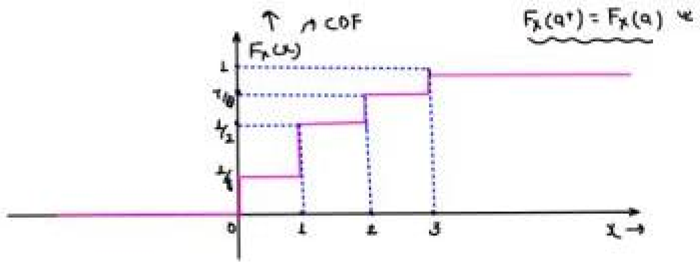
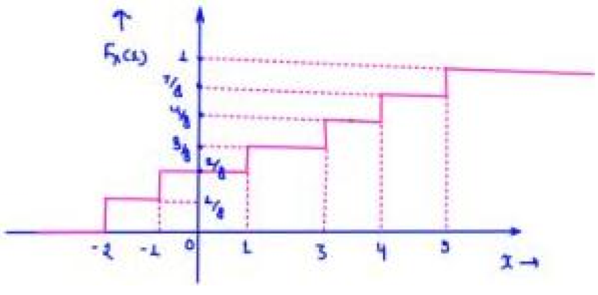


$$F_x(s) = F_y(s^+) = L$$

$$F_x(0) = F_x(0^+) = 1/8 \quad ; \quad F_x(0^-) = 0$$

$$F_x(1.33) = 1/2 = F_y(1.33^+) = F_x(1.33) \checkmark$$

Q



COF of av x is given. Find (a) P(-L<x≤4)

(b) P(-L<x<4)

(c) P(-L≤x<4)

(d) P(-L≤x≤4)

(e) P(x=2)

(f) P(x=4)

(a) P(-L<x≤4) = P(x≤4) - P(x≤L)

$$= \boxed{F_x(4) - F_x(-L)}$$

$$= \boxed{F_x(4^+) - F_x(-L^+)}$$

$$= \frac{7}{8} - \frac{2}{8} = \frac{5}{8}$$

$$F_x(x) = P[x \leq x]$$

$$F_x(a^+) = F_x(a)$$

(b) P(-L<x<4) = P(x<4) - P(x≤L)

$$= P(x \leq 4) - P(x = 4) - P(x \leq -L)$$

$$\sqrt{P(-L < x < 4) = F_x(4) - F_x(-L) - P(x = 4)}$$

$$P(x < 4) = P(x \leq 4) - P(x = 4)$$

$$F_x(4^-) = P[x \leq 4^-] = P[x < 4]$$

$$\sqrt{P(-L < x < 4) = F_x(4^-) - F_x(-L)}$$

$$\sqrt{P(-L < x < 4) = F_y(4^-) - F_x(-L^+)}$$

$$= \frac{4}{8} - \frac{2}{8} = \frac{5}{8} = \frac{1}{4}$$

(f) P(x=4) = P(x≤4) - P(x≤4⁻)

$$P(x = 4) = F_x(4) - F_x(4^-)$$

$$P(x = 4) = F_x(4^+) - F_x(4^-)$$

$$= \frac{7}{8} - \frac{4}{8} = \frac{3}{8}$$

(c) P(-L≤x<4) = P(x<4) - P(x<-L)

$$= P(x \leq 4^-) - P(x \leq -L^-)$$

$$P(-L \le x < 4) = F_x(4^-) - F_x(-L^-)$$

$$= F_x(4^+) - P(x=4) - \{F_y(-L^+) - P(x=-L)\}$$

$$P(-L \le x < 4) = F_y(4) - F_y(-L) - P(x=4) + P(x=-L)$$

$$\sqrt{4^{-4}}$$

$$P(x \le 4^-) = P(x < 4)$$

$$P(x \le a^-) = P(x < a)$$

$$\wedge F_x(4)$$

$$\vee P(x=4) = F_x(4^+) - F_x(4^-)$$

$$F_x(4^-) = F_x(4^+) - P(x=4)$$

$$F_y(-L^-) = F_x(-L^+) - P(x=-L)$$

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(d) $$\frac{P(-1 \le x \le 4) = F_x(4) - F_x(-1) + P(x=-1)}{P(-1 \le x \le 4) = F_x(4^+) - F_x(-1^-)}$$	$$-1 \quad 4$$ $$F_x(4) = P(x \le 4)$$ $$F_x(-1) = P(x \le -1)$$ $$P(x=-1) = F_x(-1) - F_x(-1^-)$$ $$P(x=-1) - F_x(-1) = -F_x(-1^-)$$	
		* Cor $\rightarrow P($ $\rightarrow$ In ca

(d) $$\frac{P(-1 \leq x \leq 4) = F_x(4) - F_x(-1) + P(x=-1)}{P(-1 \leq x \leq 4) = F_x(4^+) - F_x(-1^-)}$$	$$\begin{gathered} -1 \quad 4 \\ F_x(4) = P(x \leq 4) \\ F_x(-1) = P(x \leq -1) \\ P(x=-1) = F_x(-1) - F_x(-1^-) \\ P(x=-1) - F_x(-1) = -F_x(-1^-) \end{gathered}$$	
		$P($
* Properties of coF of a R.V. x:- (i) $$F_x(\infty) = P[x \leq \infty] = 1$$ (ii) $$F_x(-\infty) = P[x \leq -\infty] = 0$$ (iii) $$F_x(x)_{min} = 0$$; $$F_x(x)_{max} = 1$$ (iv) $$F_x(x)$$ is a non-decreasing function. (v) if $$x_2 > x_1 \Rightarrow F_x(x_2) \geq F_x(x_1)$$ (vi) $$F_x(a^+) = F_x(a) \Rightarrow$$ Costi from R.H.S. (vii) $$P(a < x \leq b) = F_x(b) - F_x(a)$$		Q: Fi

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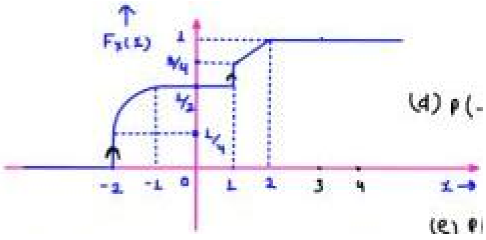
Q. COF of a R.V. X is shown

X: Mixed R.V.



Here is your text converted into a clean, structured format optimized for Obsidian. I have applied standard Obsidian markdown, formatted the math using MathJax formatting, used callout blocks for important notes, and gently corrected a few minor notation typos (like adjusting the dummy variable in the integral) to ensure your notes are mathematically perfectly sound.

Probability Distribution Functions

Definitions

- $f_X(x)$: **Probability Density Function (PDF)**
  - It tells how dense the probability is around a particular value of a continuous Random Variable (R.V.).
- $P_X(x) = P(X = x)$: **Probability Mass Function (PMF)**
- $F_X(x) = P(X \leq x)$: **Cumulative Distribution Function (CDF)**

> **Note on CDF**
>
> The CDF is **NOT** the probability at a single point.

Relation between PDF and CDF

The relationship between the PDF and CDF is given by differentiation and integration:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$$

Solved Calculations

Find:

$P(X = 1)$, $P(X = 2)$, $P(X = -2)$, $P(-1 < X \leq 3)$, $P(1 \leq X < 4)$

Solutions:

- **(a)** $P(X = 1) = F_X(1) - F_X(1^-) = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}$
- **(b)** $P(X = 2) = F_X(2) - F_X(2^-) = 1 - 1 = 0$

- **(d)** $P(-1 < X \leq 3) = F_X(3) - F_X(-1) = 1 - \frac{1}{2} = \frac{1}{2}$
- **(e)** $P(1 \leq X < 4) = F_X(4) - F_X(1^-) = 1 - \frac{1}{2} = \frac{1}{2}$
- **(f)** $P(X = -2) = F_X(-2) - F_X(-2^-) = \frac{1}{4} - 0 = \frac{1}{4}$

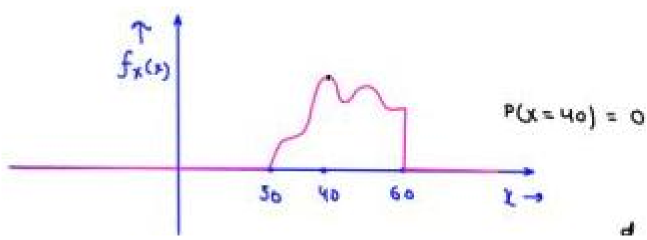
Example: Quiz Solving Time

A group of students typically takes **30 min to 60 min** to solve a Quiz.

A random variable X is defined such that it represents the time taken by a randomly selected student to solve the Quiz.

$$X \in [30, 60]$$

(The PDF of R.V. X is shown here).



COF of R.V. $\rightarrow F_X(x)$

PDF of R.V. $\rightarrow f_X(x)$

$$\frac{d}{dx} F_X(x) = f_X(x)$$

$$F_X(x) = \int_{-\infty}^{x} f_Q(q)\, dq$$

$$\frac{d}{dx} \text{COF} = \text{PDF}$$

$$\int \text{PDF}.\, dx = \text{COF}$$

$\rightarrow P[a < X \leq b] = F_X(b) - F_X(a) = F_X(b+) - F_X(a+)$

$$= \int_{-\infty}^{b} f_Q(q)\, dq - \int_{-\infty}^{a} f_Q(q)\, dq = \int_{a}^{-\infty} f_Q(q)\, dq + \int_{-\infty}^{b} f_Q(q)\, dq$$

$$P[a < X \leq b] = \int_{a+}^{b+} f_X(x)\, dx$$

$$P[a \leq X \leq b] = \int_{a-}^{b-} f_X(x)\, dx$$

$$P[a < X < b] = \int_{a}^{b-} f_X(x)\, dx$$

$$P[a \leq X < b] = \int_{a-}^{b-} f_X(x)\, dx$$

$$P[a < X \le b] = \int_a^b f_Q(q)\,dq = \int_{a+}^{b+} f_Q(q)\,dq$$

$$F_X(x) = \int_{-\infty}^{x} f_X(x)\,dx$$

$$= \int_{-\infty}^{x} f_Q(q)\,dq$$

$$P(x = a) = \int_a^{a+} f_X(x)\,dx$$

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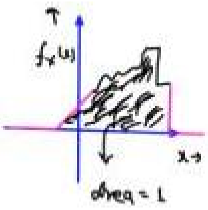
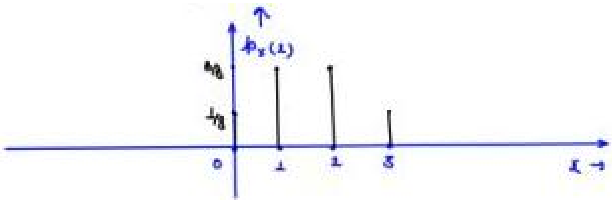
- Properties:-
  COF → 0 to L; non-decreasing
  PDF → Not negative and area under the curve = L

Since $F_x(x)$ is non-decreasing $\Rightarrow \frac{d}{dx}F_x(x) = f_x(x) \ge 0$

(i) $f_x(x) \ge 0$; PDF will never be negative



4 $p(-\infty < x \le \infty) = F_x(\infty) - F_y(-\infty) = L - 0 = L$



(ii) $\int_{-\infty}^{\infty} f_x(1)\,dx = L$; area under PDF curve = L

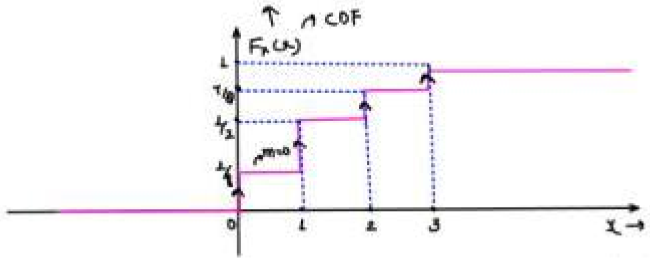
4 $p(a < x \le b) = \int_{a^+} f_x(1)\,dx =$

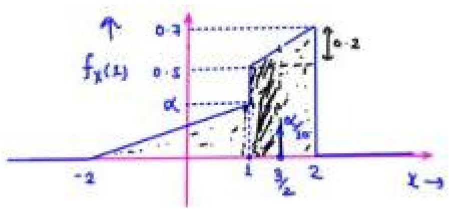
Q. x: no. of heads appearing when s coins tossed simultaneously.

Draw PMF, PDF & COF of R.V. x?

4 $x \in \{0, 1, 2, 3\} \rightarrow \mathrm{DRV}$

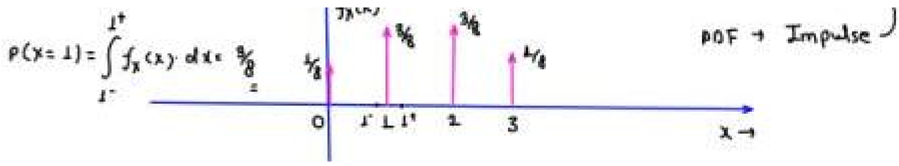
$p(x=0) = 1/8, \quad p(x=1) = 3/8, \quad p(x=2) = 3/4, \quad p(x=3) = 1/8$





PDF $f_x(1) = \frac{d}{dx} F_x(x)$

COF → sudden Imp



$p(x=1) = \int_{1^-}^{1^+} f_x(x)\, dx = 3/8$

↓
DRV
PDF → Impulse

Q.

x: Mixed R.V.

$m = \frac{0.2}{L} = 0.1 = \frac{y_2 - 0.5}{3/2 - L}$

$0.1 = \frac{y_2 - 0.5}{0.5}$

$y_2 = 0.6$

(a) find α

(a) $\alpha = ?$

(b) $p\{x = 1\}$

for PDF, area under curve = L

(c) $p\{x = 3/2\}$

$\int_{-\infty}^{\infty} f_x(1)\, dx = L$

(d) $p\{1 \le x < 3/2\}$

$= \frac{1}{2} \times 3 \times \alpha + \frac{1}{2} \times 1 \times 0.2 + 0.5 \times L + \frac{\alpha}{10} = L$

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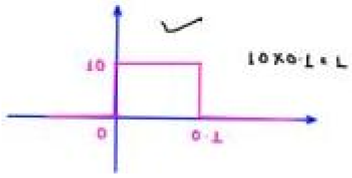
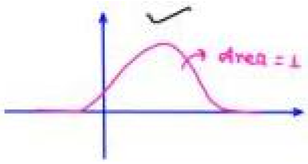
$$\Rightarrow 1.6\alpha + 0.6 = 1$$

$$\Rightarrow \alpha - \frac{0.4}{1.6} = 0.25$$

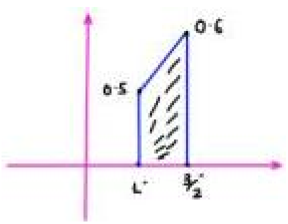
$$\alpha = 1/4$$





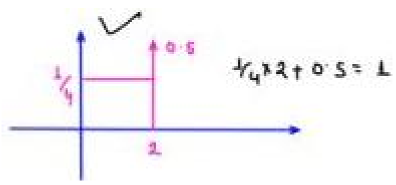
(b)

$$p(x=1) = \int_1^{1^*} f_x(x) \cdot dx = 0$$



(c)

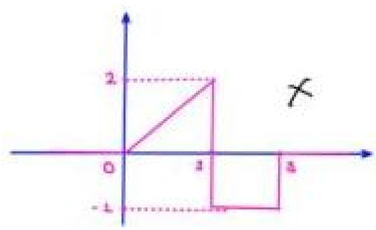
$$p(x=3/2) = \int_{3/2}^{3/2} f_x(x) \cdot dx = \frac{\alpha}{10} = \frac{1}{40}$$

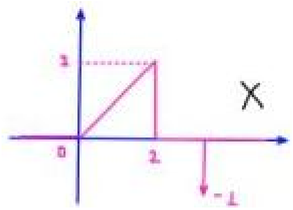


(d)

$$p(1 \leq x < 3/2) = \int_{1}^{3/2} f_x(x) \cdot dx = \frac{1}{2} \times (1.1) \times 1/2 = \frac{1.1}{4}$$

Q. Valid PDF = ?





Q. PDF of RV x is given

$$f_x(x) = p(x) = ke^{-3|x|}; \quad -\infty < x < \infty$$

Find the value of k = ?

$$\int_{-\infty}^{\infty} f_x(x) \cdot dx = \int_{-\infty}^{\infty} ke^{-3|x|} \cdot dx = 1$$

$$\Rightarrow 2 \int_{0}^{\infty} ke^{-3x} \cdot dx = 1$$

$$\Rightarrow \frac{2k}{3} = 1 \quad \Rightarrow k = 1.5$$

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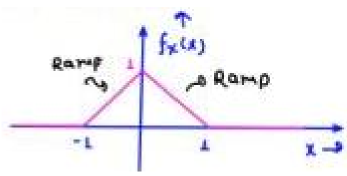
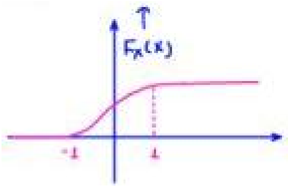
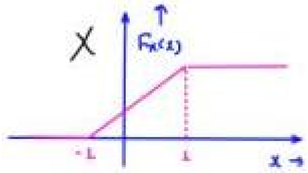
Q. PDF of RV x is shown.



$$\int_{-\infty}^{x} f_x(x) \cdot dx = F_x(x)$$

CDF = ?





Q. PDF $f_x(x) = ae^{-b|x|}$; $b > 0$, $a > 0$

$$f_q(q) = ae^{-b|q|}$$

Find cDF for $x > 0$

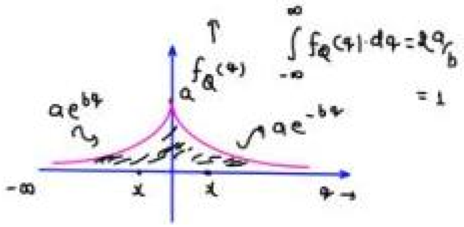
4

$$F_x(x) = \int_{-\infty}^{x} f_q(q) \cdot dq$$

$$= \int_{-\infty}^{0} f_q(q) \cdot dq + \int_{0}^{x} f_q(q) \cdot dq$$

$$= \int_{-\infty}^{0} ae^{bq} \cdot dq + \int_{0}^{x} ae^{-bq} \cdot dq$$

$$= \left[\frac{ae^{bq}}{b}\right]_{-\infty}^{0} + \left[\frac{ae^{-bq}}{-b}\right]_{0}^{x}$$



$$f_q(q) = \begin{cases} ae^{-bq} & ; x > 0 \\ ae^{bq} & ; x < 0 \end{cases}$$

$$= \frac{a}{b}[1 - e^{-\infty}] - \frac{a}{b}[e^{-bx} - 1]$$

$$= \frac{a}{b} - \frac{a}{b}[e^{-bx} - 1]$$

$$= \frac{a}{b} - \frac{a}{b}e^{-bx} + \frac{a}{b} = \frac{2a}{b} - \frac{a}{b}e^{-bx} = \frac{a}{b}[2 - e^{-bx}]$$

$$F_x(x) = \frac{a}{b}[2 - e^{-bx}] \quad ; \quad x > 0$$

$$F_x(-\infty) = 0$$

$$F_x(\infty) = \lambda a/b = 1$$

4 $x \leq 0$;

$$F_x(x) = \int_{-\infty}^{x} ae^{bx} \cdot dq = \frac{a}{b}[e^{bq}]_{-\infty}^{x} = \frac{a}{b}e^{bx} \quad ; \quad x \leq 0$$

$$F_x(x) = \frac{a}{b}e^{bx}u(-x) + \frac{a}{b}[2 - e^{-bx}]u(x)$$

Q. Find the following probabilities in terms of cDF

and PDF

(a) $p(x < 2) = p(x \leq 2) - p(x = 2) = F_x(2) - p(x = 2) = \int_{-\infty}^{2} f_x(x) \cdot dx$
(b) $p(x > 2) = 1 - p(x \leq 2) = 1 - F_x(2) = \int_{2}^{\infty} f_x(x) \cdot dx$
(c) $p(x \geq 2) = 1 - p(x \leq 2) + p(x = 2) = 1 - F_x(2) + p(x = 2) = \int_{2}^{\infty} f_x(2) \cdot dx$
(d) $p(|x| > 2) = p(x > 2) + p(x < -2)$
(e) $p(|x| \leq 2) = p(-2 \leq x \leq 2) = F_x(2) - F_x(-2) = F_x(2) - F_x(-2) + p(x = -2)$

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①

$$p(x < z) = p(-\infty < x < z) = F_x(z^-) - F_z(\infty) = F_x(z^-) = F_x(z) - p(x = z)$$

$$= \int_{-\infty}^{z^+} f_x(z)\, dz$$

②

$$p(x > z) = p(z < x \leq \infty) = F_x(\infty) - F_x(z)$$

$$= 1 - F_x(z)$$

$$p(x = z) = F_x(z) - F_x(z^-)$$

$$= \int_{z^+} f_x(z)\, dz$$

$$F_x(z^-) = F_x(z) - p(x = z)$$

③

$$p(x \geq z) = p(z \leq x \leq \infty) = F_x(\infty) - F_x(z^-) = 1 - F_x(z^-)$$

$$= \int_{z^+} f_x(z)\, dz$$

$$= 1 - F_x(z) + p(x = z)$$

Q. 4 R.v. x is called symmetrical about 0 if v xer

$$p[x \geq x] = p[x \leq -x]$$

$$F_x(x)$$

is cof of x where x is symmetrical about 0.

$$(i) \quad p\{|x| \leq x\} \text{ is } (a)1-2F_x(x) \quad (b)1+2F_x(x) \quad (c)2F_x(x)-1 \quad (d)F_x(x)-1$$

$$(ii) \quad p\{|x| > x\} \text{ is } (a)2[1-F_x(x)] \quad (b)2[1+F_x(x)] \quad (c)1-F_x(x) \quad (d)1+F_x(x)$$

$$(iii) \quad p\{x = x\} \text{ is } (a)F_x(x)-F_x(-x)-1 \quad (b)F_x(x)+F_x(-x)+1$$

$$(c)F_x(x)-F_x(-x)+1 \quad (d)F_x(x)+F_x(-x)-1$$

$$(ii) \quad p(|x| \leq x) = p(-x \leq x \leq x) = F_x(x) - F_x(-x) \quad \frac{p(x=-x) = F_x(-x) - F_x(x)}{-x-x} = F_x(x) - F_x$$

$$p[x \geq x] = p[x \leq -x]$$

$$1 - p[x < x] = p[x \leq -x]$$

$$1 - p[x \leq x] + p[x = x] = p[x \leq -x]$$

$$1 - F_x(x) + p[x = x] = F_x(-x)$$

$$(iii) \quad p[x = x] = F_x(-x) + F_x(x) - 1 = p(x = -x)$$

$$p[x = -x] = F_x(x) + F_x(-x) - 1$$

$$(ii) \quad p(|x| \leq x) = F_x(x) - F_x(-x) + p(x = -x)$$

$$= F_x(x) - F_x(-x) + F_x(x) + F_x(-x) - 1$$

$$p(|x| \leq x) = 2F_x(x) - 1$$

$$(iii) \quad p(|x| > x) = p(x > x) + p(x < -x)$$

$$= 1 - p(x \leq x) + p(x \leq -x) - p(x = -x)$$

$$= 1 - F_x(x) + F_x(-x) - F_x(x) - F_x(-x) + 1$$

$$= 2 - 2F_x(x)$$

$$p[|x| > x] = 2[1 - F_x(x)]$$

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- Some Mathematics:-
- Conditional Probability:-

OR → + → Union (U)

AND → X · → Intersection

(A)

Conditional probability is the probability of an event A occurring given that another event B has already occurred.

Q: A die is rolled. What is the probability that the number is even, given that it is greater than 2?

$$\boxed{P(\frac{A}{B}) = \frac{P(A \cap B)}{P(B)}} = \frac{2/6}{4/6} = 1/2$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{2, 4, 6\}$$

$$B = \{3, 4, 5, 6\}$$

4

$$P(A \cap B) = P(\frac{A}{B}) \cdot P(B) = P(\frac{B}{A}) \cdot P(A)$$

$$P(A/B) = \frac{P(A \cap B)}{P(A)}$$

$$P(B/A) = \frac{P(B \cap A)}{P(A)}$$

4

If A and B are independent events.

$$\boxed{P(A/B) = P(A)}$$

$$P(A \cap B) = P(A/B) \cdot P(B)$$

$$\boxed{P(A \cap B) = P(A) \cdot P(B)}$$

Eq. 4 You roll 2 dice SI mutatenously.

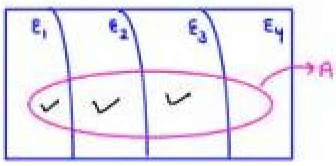
✓ A: "First die shows an even number"

✓ B: "Second die shows a 3"

$$P(A) = \frac{3}{6} = 1/2 \quad P(B) = 1/6$$

$$P(A/B) = P(A) = 1/2$$

$$P(B/A) = P(B) = 1/6$$



$$\varepsilon_1 \quad \varepsilon_2 \quad \varepsilon_3 \quad \varepsilon_4$$

$$P(A) = P(\varepsilon_1 \cap A) + P(\varepsilon_2 \cap A) + P(\varepsilon_3 \cap A) + P(\varepsilon_4 \cap A)$$

$$P(A) = P(\varepsilon_1) \cdot P(\frac{A}{\varepsilon_1}) + P(\varepsilon_2) \cdot P(\frac{A}{\varepsilon_2}) + P(\varepsilon_3) \cdot P(\frac{A}{\varepsilon_3}) + P(\varepsilon_4) \cdot P(\frac{A}{\varepsilon_4})$$

Q

Suppose:

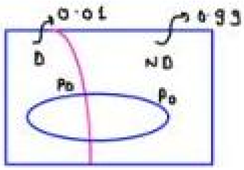
- 1% of people have a disease: $P(D) = 0.01$, $P(\text{No } D) = 0.99$
- Test is 99% accurate:
- $P(\text{Positive} \mid D) = 0.99$
- $P(\text{Positive} \mid \text{No } D) = 0.01$

Q: What is the probability that a person actually has the disease given their test is positive?

$$P\left(\frac{D}{P_0}\right) = \frac{P(D \cap P_0)}{P(P_0)} = \frac{0.99 \times 0.01}{P(P_0)}$$

$$P\left(\frac{P_0}{D}\right) = \frac{P(P_0 \cap D)}{P(D)} = 0.99$$

$$\rightarrow P(D \cap P_0) = 0.99 \times 0.01$$



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$$P(P_0) = P(0 \cap P_0) + P(\omega_0 \cap P_0)$$

$$P(P_0) = P(0) \cdot P\left(\frac{P_0}{0}\right) + P(\omega_0) \cdot P\left(\frac{P_0}{\omega_0}\right)$$

$$= 0.01 \times 0.99 + 0.99 \times 0.01$$

$$= 2 \times 0.01 \times 0.99$$

$$P\left(\frac{D}{P_0}\right) = \frac{0.99 \times 0.01}{2 \times 0.01 \times 0.99} = \frac{1}{2} = \boxed{0.5}$$

Q. COF of RV Y is given as:-

$$F_Y(y) = (1 - e^{-y})U(y)$$

$$P\left[\frac{Y < 2}{Y > -1}\right] = P$$

$$F_Y(-1) = (1 - e^{-1})U(y)$$

$$= 0$$

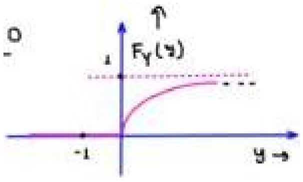
$$F_Y(2) = (1 - e^{-2})U(y)$$

$$= 1 - e^{-2}$$

$$P\left[\frac{Y<2}{Y>-1}\right] = \frac{P[(Y<2)\cap(Y>-1)]}{P[Y>-1]} = \frac{P[-1<Y<2]}{P[Y>-1]} = \frac{F_Y(2)-F_Y(-1)}{1-P[Y\le -1]}$$

$$P\left[\frac{Y<2}{Y>-1}\right] = \frac{F_Y(2)-F_Y(-1)}{1-F_Y(-1)} = \frac{1-e^{-2}-0}{1-0}$$

$$\boxed{Ans \to 1-e^{-2} \quad Ans.}$$



Q. let sample space

$$s = \{s_i\}_{i=1}^{4}$$

$$P(s_1) = \frac{1}{6}, \quad P(s_2) = \frac{1}{3}, \quad P(s_3) = \frac{1}{3}, \quad P(s_4) = \frac{1}{6}$$

4. R. r.

$$X(s_i)$$

is defined such that

$$X(s_i) = i - 2$$

;

$$1 \le i \le 4$$

Draw PMF

$$s = \{s_1, s_2, s_3, s_4\}$$

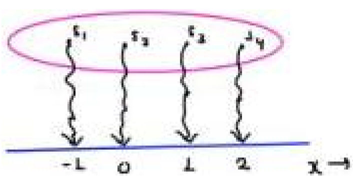
$$x \in \{-1, 0, 1, 2\}$$

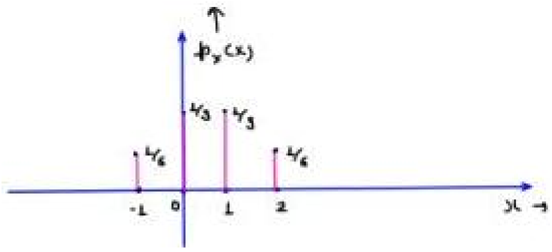
$$P(x=-1) = P(s_1) = \frac{1}{6}$$

$$P(x=0) = P(s_2) = \frac{1}{3}$$

$$P(x=1) = P(s_3) = \frac{1}{3}$$

$$x \in \{-1\} \text{ when } i = 1$$





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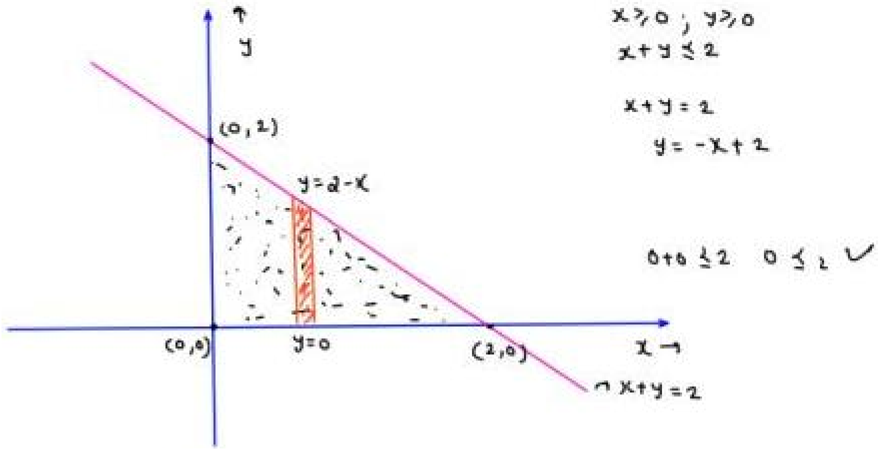
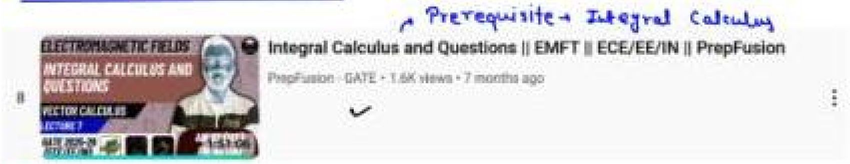


Some Basic Mathematics:-

↗ Prerequisite → Integral Calculus



Q. Draw region R.

R: $x > 0$; $y > 0$; $x + y \leq 2$

Find

$$\iint_R dx \cdot dy$$

,

$$\iint_R xy \, dx \, dy$$

①

$$\iint_R dx \, dy = \text{drea} = 1/2 \times 2 \times 2 = 2$$

②

$$\iint_R xy\,dx\,dy = \int_{x=0}^{2}\int_{y=0}^{y=2-x} xy\,dx\,dy$$

$$= \int_{x=0}^{2} x\cdot dx \int_{y=0}^{y=2-x} y\,dy = \frac{1}{2}\int_{x=0}^{2} x(x^2+4-4x)\cdot dx$$

$$= \underbrace{\left[\frac{y^2}{2}\right]_0^{2-x}}_{=\infty} = \frac{1}{2}\int_0^2 x^3-4x^2+4x\cdot dx$$

$$= \int_{x=0}^{2} x\cdot\frac{(x-2)^2}{2}\cdot dx = \frac{1}{2}\left[\frac{x^4}{4}-\frac{4x^3}{3}+2x^2\right]_0^2$$

$$= \frac{1}{2}[4-\frac{3}{2}+8] = \frac{2}{3}\ \mathrm{dA}$$

Q. Draw region R'.

R': x > 0 ; y > 0 ; x + y ≤ 2 ; x + y ≥ 1

Find

$$\iint_R dx\cdot dy$$

,

$$\iint_R xy\,dx\,dy$$

①

$$\iint_{R'} dx\cdot dy = \frac{1}{2}x^2+2-\frac{1}{2}x^1+1$$

$$= 2-1/2 = 3/2 = 1.5\ \mathrm{w}$$

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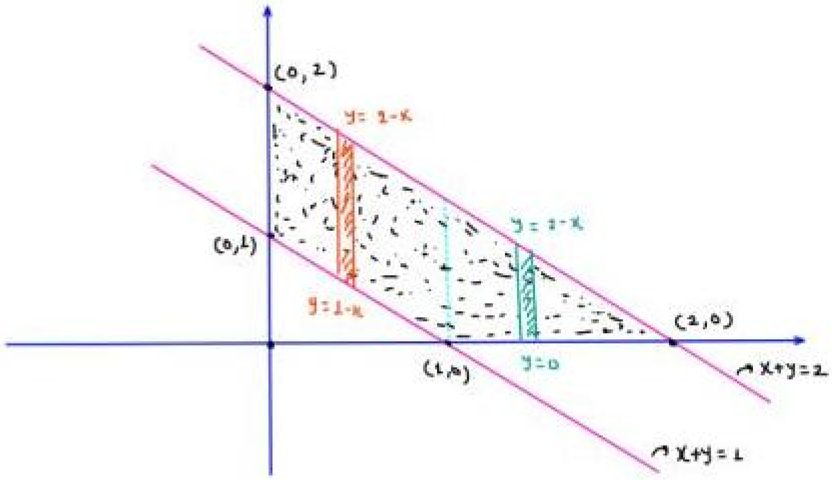
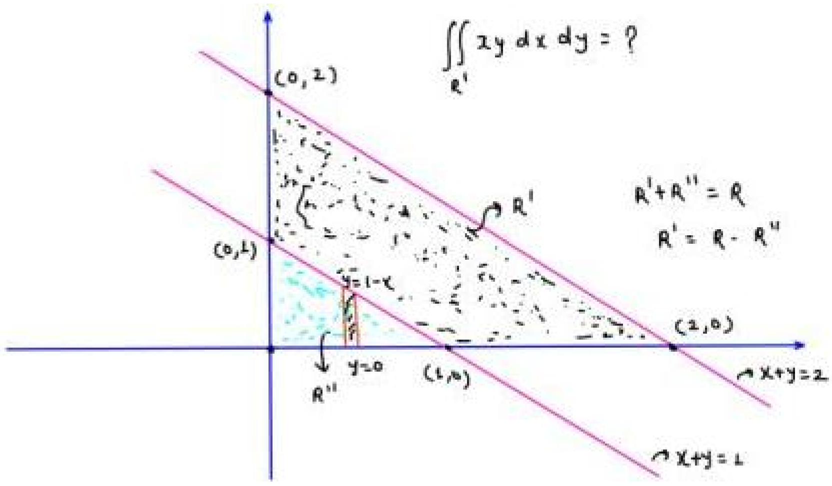


$$\textcircled{R} \iint_{R^1} xy\,dx\,dy = \int_{x=0}^{x=1}\int_{y=1-x}^{2-x} xy\,dx\,dy + \int_{x=1}^{2}\int_{y=0}^{2-x} xy\,dx\,dy$$

$$= \int_{x=0}^{x=1} x\left[\frac{y^2}{2}\right]_{1-x}^{2-x} \cdot dx + \int_{x=1}^{2} x\left[\frac{y^2}{2}\right]_{0}^{2-x} \cdot dx$$



\begin{aligned} \iint_{R^1} xy \, dx \, dy &= \iint_{R} xy \, dx \, dy - \iint_{R"} xy \, dx \, dy \quad x^2 +

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- 2-dimensional Random Variable:-

Let two R.v. x and y are given.

4 Random Variable x:-

$$\int_{\text{cof of R.v. } x}^{F_x(x)=\rho(x\le x)} f_x(x) = \frac{d}{dx}F_x(x) \quad ; \quad F_x(x) = \int_{-\infty}^{x} f_{\mathbb{R}}(x)\,dx$$

5 Random Variable y:-

$$\int_{\text{cof of R.v. } y}^{F_y(y)=\rho(y\le y)} f_y(y) = \frac{d}{dy}F_y(y) \quad ; \quad F_y(y) = \int_{-\infty}^{y} f_{\mathbb{R}}(y)\,dy$$

4 Joint cof of Random Variable x and y:-

$$F_{xy}(x, y) = \rho[x \le x \cap y \le y] = \rho[x \le x, y \le y]$$

$$F_{xy}(2, 3) = \rho[x \le 2, y \le 3]$$

Joint cof of R.v. x and y:-

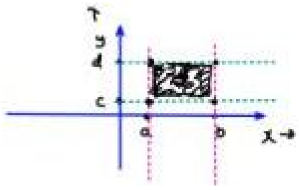
$$\frac{\partial^2 F_{xy}(x, y)}{\partial x \partial y} = f_{xy}(x, y)$$

$$F_{xy}(1, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{PQ}(P, q)\, dp\, dq$$

- Properties of Joint cof:-

$$\checkmark ① F_{xy}(x, \infty) = \rho[x \le x, y \le \infty] = F_x(x) \rightarrow \text{Marginal cof of } R.v.x$$

$$\checkmark ② F_{xy}(\infty, y) = \rho[x \le \infty, y \le y] = F_y(y) \rightarrow \text{Marginal cof of } R.v.y$$



$$③ F_{xy}(\infty, \infty) = \rho[x \le \infty, y \le \infty] = 1$$

$$④ F_{xy}(x, -\infty) = \rho[x \le x, y \le -\infty] = 0$$

$$⑤ F_{xy}(-\infty, y) = \rho[x \le -\infty, y \le y] = 0$$

$$⑥ F_{xy}(-\infty, -\infty) = \rho[x \le -\infty, y \le -\infty] = 0$$

⑦ Non-decreasing function:-

$$x_2 \ge x_1, y_2 \ge y_1$$

$$4F_{xy}(x_2, y_2) \ge F_{xy}(x_1, y_1)$$

⑧ Bounded b/w 0 to 1

$$0 \le F_{xy}(x, y) \le 1$$

⑨ Probability of a rectangle:-

$$\rho[a < x \le b, c < y \le d] = F_{xy}(b^2, d^2) + F_{xy}(a^2, c^2) - F_{xy}(a^2, d^2) - F_{xy}(b^2, c^2)$$

$$= F_{xy}(b, d) + F_{xy}(a, c) - F_{xy}(a, d) - F_{xy}(b, c)$$

$$F_x(x) \rightarrow \text{cof} \checkmark$$

$$F_y(y) \rightarrow \text{cof}$$

$$F_{xy}(x, y) \rightarrow \text{cof} \checkmark$$

$$\rho[a < x \le b] = F_x(b) - F_x(a)$$

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Q. x and y are two R-V. such that
x ∈ {0, 1, 2} ~ORV; y ∈ {1, 2, 3} ~ORV

we define their Joint PMF P(x=x, y=y) as:

	L y=1	y=2	y=3
→x=0	0.15	0.05	0.1
→x=1	0.1	0.03	0.17
→x=2	0.05	0.23	0.1

Find P(0<x≤2, 1<y≤3) = ?
Prob → x ∈ {1,2}, y ∈ {2,3}

$$p_x(3) = P(x = x)$$

$$p_y(3) = P(y = y)$$

$$p_{xy}(3,3) = P(x = x, y = y)$$

$$P(x = 0, y = 1) = 0.15$$

$$P(x = 1, y = 3) = 0.17$$

$$P(x = 3, y = 2) = 0$$

$$\underline{\underline{M-1}} \quad P[0 < x \leq 2, 1 < y \leq 3] = P[x = 1, y = 2] + P[x = 1, y = 3]$$

$$+P[x = 2, y = 2] + P[x = 2, y = 3]$$

$$= 0.03 + 0.17 + 0.25 + 0.1$$

$$= \boxed{0.55}_{\text{Ans.}}$$

$$\underline{\underline{M-2}} \quad P[0 < x \leq 2, 1 < y \leq 3] = F_{xy}(2,3) + F_{xy}(0,1) - F_{xy}(0,3) - F_{xy}(1,1)$$

$$= 1 + 0.15 - 0.3 - 0.3 = \boxed{0.55}_{\text{Ans.}}$$

$$F_{xy}(2,3) = P[x \leq 2, y \leq 3] = 1; \quad F_{xy}(0,1) = P[x \leq 0, y \leq 1] = 0.15$$

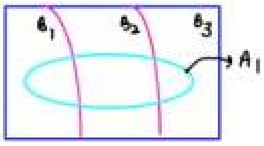
$$F_{xy}(0,3) = P[x \leq 0, y \leq 3] = 0.3; \quad F_{xy}(2,1) = P[x \leq 2, y \leq 1] = 0.3$$

$$\Rightarrow P(x = 0) = P(x = 0, y = 1) + P(x = 0, y = 2) + P(x = 0, y = 3)$$

$$= 0.15 + 0.05 + 0.1 = 0.3$$

	B_{1}	B_{2}	B_{3}
A_{1}	A*{1}∩B*{1}	A*{1}∩B*{2}	A*{1}∩B*{3}
A_{2}			
A_{3}			

$$P(A_1) = P(A_1 \cap A_1) + P(A_2 \cap A_2)$$
$$+P(A_3 \cap A_3)$$



$$\Rightarrow P(y=2) = P(x=0, y=2) + P(x=1, y=2) +$$
$$P(x=2, y=2)$$
$$= 0.05 + 0.03 + 0.25 = 0.33 = \text{Ans.}$$

- Properties of Joint PDF:-

① Joint PDF of x and y from Joint COF:-

$$f_{xy}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{xy}(x,y)$$

② Getting marginal PDF of x and y:-

→ Joint COF

$$\text{Marginal PDF of } x: \quad f_x(x) = \frac{d}{dx} F_x(x); \quad F_{xy}(x,\infty) = F_x(x)$$
$$\hookrightarrow \text{marginal COF of } x$$
$$\text{marginal} \underset{\text{PDF of } x}{\rightarrow} \underbrace{f_x(x)}_{\text{function of } x} = \int_{y=\infty}^{\infty} f_{xy}(x,y) \cdot dy$$

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Marginal PDF of Y:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$
$$F_{XY}(\infty, y) = F_Y(y)$$
$$f_Y(y) = \int_{x=-\infty}^{\infty} f_{XY}(x,y) \cdot dx$$

③ Non - Negative:-

$$f_X(x) \geq 0, \quad f_Y(y) \geq 0; \quad f_{XY}(x,y) \geq 0$$
$$f_X(x) \rightarrow \text{POF}$$
$$f_Y(y) \rightarrow \text{POF}$$

$$f_{XY}(x,y) \to \text{POF}$$

④

$$\int_{-\infty}^{\infty} f_X(x) \cdot dx = \int_{-\infty}^{\infty} f_Y(y) \cdot dy = \iint_R f_{XY}(x,y)\, dx\, dy = L$$

⑤ If x and Y are independent R.V. then

$$F_{XY}(x,y) = F_X(x) \cdot F_Y(y)$$

$$f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$$

Joint PDF/CBF of R.V. = [Marginal PDF/CBF of R.V. x] x [Marginal PDF/CBF of R.V. Y]

If A and B are independent events

$$P(A \cap B) = P(A) \cdot P(B)$$

②

$$f_{XY}(x,y) = \begin{cases} be^{-(x+y)} & ;\ 0 < x < 2 \text{ and } 0 < y < \infty \\ 0 & ;\ 0/\infty \end{cases}$$

(i) find b = ?

(ii)

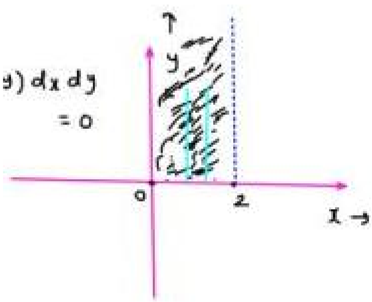
$$P(X+Y<L) = ?$$

(iii)

$$P(X+Y=L) = ? = 0 = \iint_{R=\text{Line}} f_{XY}(x,y)\, dx\, dy = 0$$

$$\iint_R f_{XY}(x,y)\, dx\, dy = L$$

$$\int_{x=0}^{2} \int_{y=0}^{\infty} be^{-x} e^{-y} \cdot dx\, dy = L$$



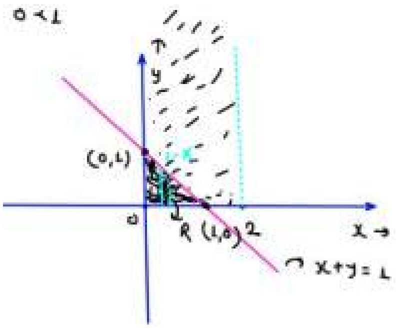
$$b \int_{x=0}^{2} e^{-x} \cdot dx \int_{y=0}^{\infty} e^{-y}\, dy = L$$

$$-b[e^{-x}]_0^2 = L$$

$$b[L \cdot e^{-L}] = L$$

$$\Rightarrow b = \frac{L}{L \cdot e^{-L}}$$

$$P(X+Y<L)=?$$



$$\iint_R f_{XY}(x,y) \cdot dx\, dy = \checkmark$$

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$$\int_{x=0}^{L} \int_{y=0}^{y=1-x} be^{-(x+y)}\, dx\, dy =$$

$$\int_{x=0}^{L} \int_{y=0}^{y=1-x} \frac{1}{1-e^{-x}} e^{-x} e^{-y}\, dx\, dy = \frac{1}{1-e^{-x}} \int_{x=0}^{L} e^{-x}\, dx \left[e^{-y}\right]_{1-x}^{0}$$

$$= \frac{1}{1-e^{-x}} \left[e^{-x} + e^{-x} x\right]_{1}^{0} = \frac{1}{1-e^{-x}} \int_{x=0}^{L} e^{-x} \left[1 - e^{x-L}\right] \cdot dx$$

$$= \frac{1}{1-e^{-x}} \left[1 - 2e^{-x}\right] = \boxed{\frac{1-2e^{-x}}{1-e^{-x}}} = \frac{1}{1-e^{-x}} \int_{x=0}^{L} \left[e^{-x} - e^{-x}\right] \cdot dx$$

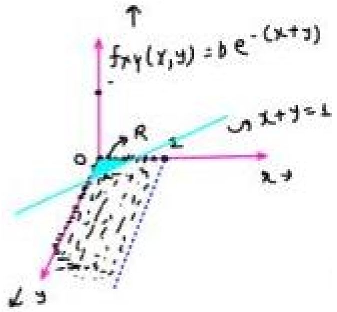
$$f_x(L) = \begin{cases} be^{-x} & ; x > 0 \\ 0 & ; 0/\omega \end{cases}$$

$$\iint_R f_{xy}(x,y)\, dx\, dy$$

$$P(X<L) = \int_{-\infty}^{L} f_x(x)\, dx$$

$$= \int_0^L be^{-x}\, dx$$

$$f_x(L) = \begin{cases} 0 & ; x < 0 \\ 0 & ; x \to 0 \end{cases}$$

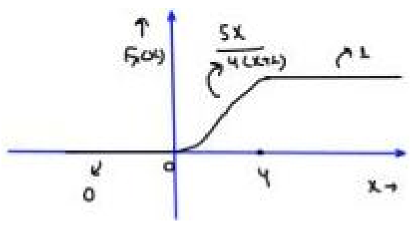


$$\int_{-\infty}^{\infty} f_x(L)\,dx = L; \int_{0}^{\infty} be^{-x}\,dx = L$$

$$b = L$$

Q. Joint cof of x and y:

$$F_{xy}(x,y) = \begin{cases} \frac{3}{4}\left[\frac{x+e^{-(x+1)y^2}}{x+1} - e^{-y^2}\right]; & 0 \le x \le 4; \\ 0 & ; x < 0 \text{ and } y < 0 \\ 1 + \frac{1}{4}e^{-5y^2} - \frac{5}{4}e^{-y^2} & ; x \ge 4 \text{ and } y \ge 0 \end{cases}$$



Find $p[2 < x \le 3]$; $p[2 < x \le 3, 1 < y \le 4]$

$$\{\text{th } \ldots\ldots\ 1\}$$

$$(9) \quad p[2 < x \le 3] = F_x(3) - F_x(2)$$

$$F_x(x) = ?$$

$$F_x(x) = F_{xY}(x,\infty) = F_{xY}(x, y=\infty)$$

$$F_x(x) = \begin{cases} \frac{5x}{4(x+1)} & ; 0 \le x \le 4 \\ 0 & ; x < 0 \\ 1 & ; x \ge 4 \end{cases}$$

$$P(2 < x \le 3) = F_y(3) - F_x(2)$$

$$= \frac{15}{16} - \frac{10}{12}$$

$$\boxed{P(2 < y \le 3) = 0.104}$$

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$$p[2 < x \le 3, L < y \le 4] = \underbrace{F_{xy}(3,4)}_{-F_{xy}(3,L)} + F_{xy}(2,L) - F_{xy}(2,4)$$

$$F_{xy}(3,4) = \tfrac{5}{4}\left[\tfrac{3+e^{-64}}{4}\right] - e^{-16}$$

$$F_{xy}(2,L) = \tfrac{5}{4}\left[\tfrac{2+e^{-3}}{3}\right] - e^{-1}$$

$$\begin{gathered}\vdots \\ \downarrow \\ \text{long calculations}\end{gathered}$$

- Conditional PDF:-

$$p\left(\frac{A}{B}\right) = \frac{p(A \cap B)}{p(B)}$$

$$f_{\frac{x}{y}}\left(\frac{x}{y}\right) = \frac{f_{xy}(1,y)}{f_y(y)} = \frac{\text{Joint PDF of } x \text{ and } y}{\text{Marginal PDF of } y}$$

$$f_{\frac{y}{x}}\left(\frac{y}{x}\right) = \frac{f_{xy}(x,y)}{f_x(x)}$$

A and B are id

$$p(A/B) = p(A)$$

$$p(B/A) = p(B)$$

$$f_{xy}(x,y) = f_{xy}(x/y) \cdot f_y(y) = f_{\frac{y}{x}}\left(\frac{y}{x}\right) \cdot f_x(x)$$

x and y are independent

$$f_{\frac{x}{y}}\left(\frac{x}{y}\right) = f_x(x) \quad ; \quad f_{\frac{y}{x}}\left(\frac{y}{x}\right) = f_y(y)$$

Q.

The joint pdf of X and Y is given by

$$0 \leq x \leq \infty$$

$$f_{xy}(x,y) = xye^{-(x^2+y^2)/2}u(x)u(y)$$

(a) Find the marginal pdf's (f{x}(x)) and (f{y}(y)), (f{x,y}(x,y), f{y,x}(x,y)).
(b) Are (X) and (Y) independent?

4

$$(9) \quad f_x(x) = \int_{y=-\infty}^{y=\infty} f_{xy}(1,y)\,dy$$

$$y = -\infty$$

$$= \int_{y=0}^{y=\infty} xye^{-(x^2+y^2)/2}\,dy$$

$$= 1e^{-x^2/2}\int_0^{\infty} ye^{-y^2/2} \cdot dy$$

$$f_{xy}(x,y) = \begin{cases} 2ye^{-(x^2+y^2)/2} & ; x > 0 \\ 0 & ; 0/\infty \end{cases}$$

$$f_x(x) = xe^{-x^2/2} \int_0^\infty ye^{-y^2/2} \cdot dy$$

$$y^2/2 = t \quad \Rightarrow \quad y\,dy = dt$$

$$y = 0 \to t = 0$$

$$y = \infty \to t = \infty$$

$$f_x(x) = xe^{-x^2/2} \int_0^\infty e^{-t} \cdot dt$$

$$\boxed{f_x(1) = xe^{-x^2/2} \cdot u(x)}$$

$$f_y(y) = \int_{x=-\infty}^{\infty} f_{xy}(x,y)\,dx$$

$$f_y(y) = \int_0^\infty xye^{-(x^2+y^2)/2}\,dx$$

$$= ye^{-y^2/2} \int_0^\infty xe^{-x^2/2} \cdot dx$$

$$\boxed{f_y(y) = ye^{-y^2/2} \cdot u(y)}$$

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$$f_{x_y}(\frac{x}{y}) = \frac{f_{xy}(x,y)}{f_y(y)} = \frac{xye^{-x^{1/2}}e^{-y^{1/2}}u(x)u(y)}{ye^{-y^{1/2}}u(y)}$$

$$\boxed{\frac{f_x(\frac{x}{y})}{f_y(\frac{x}{y})} = xe^{-x^{1/2}}u(x) = f_y(x)}$$

$$\boxed{\frac{f_y(\frac{y}{2})}{f_x(\frac{y}{2})} = ye^{-y^{1/2}}u(y) = f_y(y)}$$

x and y are independent.

(b) x and y are independent? → YES

$$f_{xy}(x,y) = f_x(x) \cdot f_y(y) \to$$

$$xye^{-x^{1/2}}e^{-y^{1/2}}u(x)u(y) = xe^{-x^{1/2}}u(x) \cdot ye^{-y^{1/2}}u(y)$$

Q

Let the joint probability density function (PDF) of random variables X and Y be given by:

$$f_{X,Y}(x,y) = \begin{cases} \text{key} & 0 < x < 1, 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$

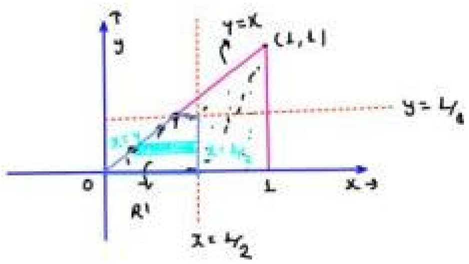
1. Find the value of K = ?
2. Compute the conditional probability:

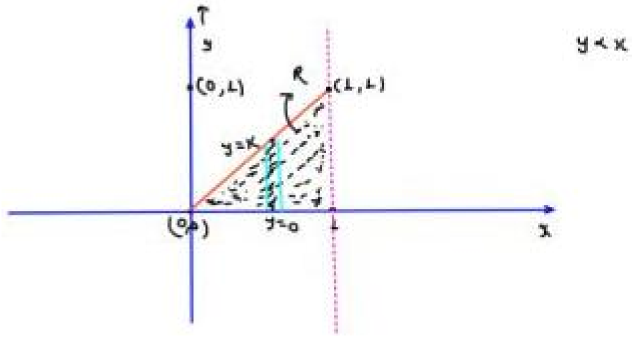
$$P\left(y < \frac{1}{8} \mid x < \frac{1}{2}\right)$$

4

$$2. \quad P\left(\frac{y < \frac{1}{8}}{x < \frac{1}{2}}\right) = \frac{P\left[(y < \frac{1}{8}), (x < \frac{1}{2})\right]}{P\left[x < \frac{1}{2}\right]} \rightarrow \text{Joint PDF} \rightarrow \text{Marginal PDF}$$

$$P\left(x < \frac{1}{2}\right) = \int_0^{\frac{1}{2}} f_y(x)\, dx$$





$$1 = \iint_R f_{xy}(x,y)\, dx\, dy = \int_{x=0}^{x+1} \int_{y=0}^{y=K} kxy\, dx\, dy = k \int_0^1 x\, dx \left[y^2_{\frac{1}{2}}\right]_0^x = k \int_0^{\frac{1}{2}} \frac{x^3}{x}\, dx$$

$$\boxed{k = 8} \quad \geqslant \frac{k}{8}\left[x^4\right]_0^1 = 1$$

$$P\left[(y < \frac{1}{8}) \cap (x < \frac{1}{2})\right] = \frac{3L}{40g\epsilon}$$

$$P\left[(y < \frac{1}{8}) \cap (x < \frac{1}{2})\right]$$

$$= \iint_{R^1} f_{xy}(x,y)\, dx\, dy$$

$$= \int_{y=0}^{\frac{1}{8}} \int_{x=y}^{x+\frac{1}{2}} 8xy\, dx\, dy$$

$$= \int_{y=0}^{\frac{1}{8}} 8y\,dy \left[\frac{x^2}{x}\right]_y^{\frac{1}{2}}$$

$$= \int_0^{\frac{1}{8}} 9 - 4y^3 \cdot dy = \left[\frac{y^2}{2} - y^4\right]_0^{\frac{1}{8}} = \frac{1}{128} - \frac{1}{64} = \frac{3L}{40g\epsilon}$$

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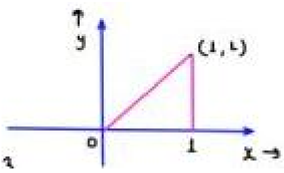
$$\begin{aligned} f_x(x) &= \int_{y=0}^{x} f_{xy}(x,y)\,dy \\ &= \int_{y=0}^{x} 8xy\,dy = 8x\left[\frac{y^2}{2}\right]_0^x = 4x^3 \end{aligned}$$

$$\rho[x < \frac{1}{2}] = \int_0^{1/2} 4x^3 \cdot dx = [x^4]_0^{1/2} = \frac{1}{16}$$

$$\boxed{\mathrm{wt}_{\mathrm{bs}} = \frac{3\,\frac{1}{4096}}{\frac{1}{16}} = 0.121} \checkmark$$



Q.

$$f_{xy} = cx(x-y)$$

;

$$0 < x < 2$$

;

$$-x < y < x$$

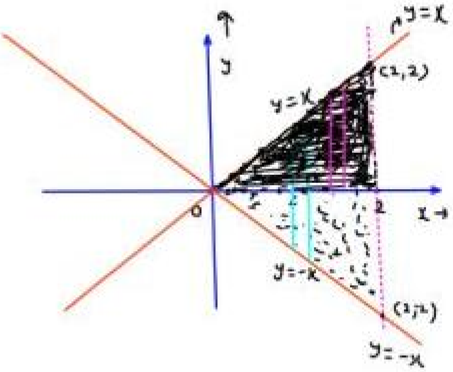
Find

$$\rho[0 < \gamma < 2] = ?$$

$$\rho[0 < \gamma < 2] = \int_0^2 f_y(y) \cdot dx$$

$$\iint_R f_{xy}(x,y)\,dx\,dy = L$$

$$\int_{x=0}^{2} \int_{y=-x}^{y=x} \underbrace{cx(x-y)}_{\downarrow \quad cx^2 - cxy} dx\,dy = L$$



$$\int_{x=0}^{2}\int_{y=-x}^{y=x} cx^2\,dx\,dy - \int_{x=0}^{2}\int_{y=-x}^{x} cxy\,dx\,dy = L$$

$$\left[\frac{2cx^4}{4}\right]_0^2 - \int_{x=0}^{2} cx\,dx\left[\frac{y^2}{2}\right]_{y=-x}^{0} dx = L$$

$$f_{xy}(x,y) = \frac{1}{8}x(x-y)$$

$$f_y(y) = \int_{x=0}^{2} f_{xy}(x,y)\cdot dx$$

$$c = 1/8$$

$$c = 1/8$$

- (i)

  $$x \to y$$

  to 2 ;

  $$y \to 0$$

  to 2
- (ii)

  $$x \to -y$$

  to 2 ;

  $$y \to -1$$

  to 0

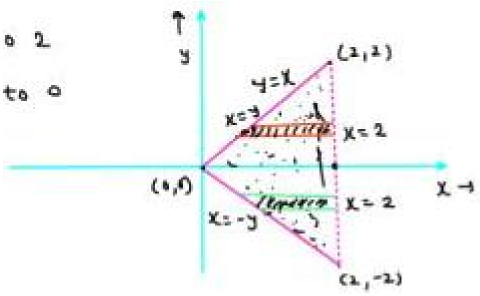
$$y \to 0 \text{ to } 2$$

$$f_y(y) = \int_{x=y}^{2} f_{xy}(x,y)\,dx$$

$$y \to -1 \text{ to } 0$$

$$f_y(y) = \int_{x=-y}^{2} f_{xy}(x,y)\cdot dx$$

$$\rho(0 < \gamma < 2) = \int_0^2 f_y(y)\,dy = ?$$



$$y = -1 \text{ to } +2$$

$$f_y(y) = \begin{cases} \sim & ; \quad 0 \le y \le 2 \\ \sim & ; \quad -2 \le y \le 0 \\ 0 & ; \quad y \ge 2; y \le 1 \end{cases}$$

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$$f_Y(y) = \int_{x=y}^{2} \frac{1}{8} x(x-y)\,dx = \frac{1}{8}\int_{x=y}^{2} (x^2 - xy)\,dx = \frac{1}{8}\left[\frac{x^3}{3} - \frac{x^2 y}{2}\right]_{x=y}^{x=1}$$

$$= \frac{1}{8}\left[\frac{8}{3} - 2y - \frac{y^3}{3} + \frac{y^2}{2}\right]$$

$$P[0 < y < 1] = \int_{y=0}^{2} \frac{1}{3} - \frac{y}{4} + \frac{y^3}{48}\,dy$$

$$f_Y(y) = \frac{1}{3} - \frac{y}{4} + \frac{y^3}{48}; 0 \le y \le 2$$

$$= \left[\frac{y}{3} - \frac{y^2}{8} + \frac{y^4}{48 \times 4}\right]_0^2 = \frac{8}{3} - \frac{1}{2} + \frac{1}{12} = \frac{8-4+1}{12} = \boxed{\frac{1}{4}}$$

$$P[0 < y < 2] = \int_{x=0}^{2}\int_{y=0}^{x} f_{XY}(x,y)\,dx\,dy = 1/4$$

Q:

Consider the joint PDF of two random variables X and Y, given by:

$$f_{X,Y}(x,y) = \begin{cases} 8 \cdot xy, & 0 \le x \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

determine the conditional PDF

$$f_{Y|X}(y|x = \frac{1}{2}).$$

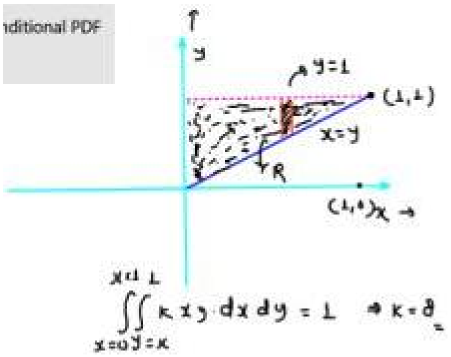
$$f_{\frac{Y}{X}}(\frac{y}{x}) = \frac{f_{XY}(x,y)}{f_X(x)}$$

$$= \frac{f_{XY}(x = 1/2, y)}{f_X(x = 1/2)}$$

$$0 \leq x$$

$$y \leq 1$$

$$x \leq y$$



$$f_X(1) = \int_{y=x}^{y=1} f_{XY}(x, y)\, dy$$

$$= \int_{y=x}^{y=1} 8xy\, dy = 4x[y^2]_x^1 = 4x[1 - x^2] = f_X(x)$$

$$f_X(x = 1/2) = 4x^{1/2}[1 - 1/4]$$

$$f_{XY}(x = 1/2, y) = 8x^{1/2} \times y = 4y$$

$$= 2 \times 8/4 = 1.5$$

$$f_{\frac{Y}{X}}(\frac{y}{x = 1/2}) = \frac{4y}{3/2} = \boxed{\frac{8y}{3}}$$

- Joint PMF:

$$p_X(x) = P(x = x)$$

$$p_{XY}(x, y) = P[x = x, y = y]$$

- Marginal PMF:

$$p_X(x) = P[x = x] = \sum_i p_{XY}(x, y_i)$$

$$p_Y(y) = P[y = y] = \sum_j p_{XY}(x_j, y)$$

- Conditional PMF:

$$p_{\frac{X}{Y}}(\frac{x}{y}) = \frac{p_{XY}(x, y)}{p_X(x)}$$

$$; \quad p_{\frac{Y}{X}}(y) = \frac{p_{XY}(x, y)}{p_Y(x)}$$

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Q. The joint PMF is given as;

$$P(x,y) = K(2x+3y) \quad ; \quad x \in \{0,1,2\}$$
$$y \in \{1,2,3\}$$

Find

(i) (P(y = 1)); (P(y = 2)); (P(y = 3)); (P(x = 0)); (P(x = 1)); (P(x = 2))
(ii) (P\left(\frac{x > 2}{y \leq 1}\right)); (P(x + y > 3)); (P(x = 1, y = 3) = \frac{11}{72})

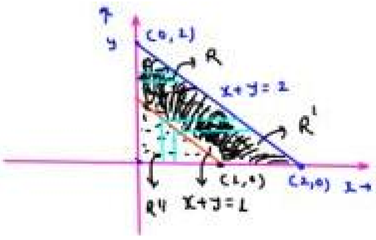
$$P(x,y) = P[x = x, y = y]$$
$$P(1,2) = P[x = 1, y = 2] = K(2+6) = 8K$$

	x	**1**	**2**	**3**	$P(x+y>3) = \frac{34}{72}$
$x \in \{0,1,2\}$	0	3K	6K	9K	
$y \in \{1,2,3\}$	1	5K	8K	11K	$P(x,y) \to PMF$
	2	7K	10K	13K	$P(x,y) = P[x=x, y=y]$
					$= K(2x+3y)$

$$3K + 6K + 9K + 5K + 8K + 11K + 7K + 10K + 13K = 1$$
$$72K = 1 \Rightarrow K = \frac{1}{72}$$
$$P(Y=1) = 15K + \frac{15}{72} = \frac{5}{24}$$
$$P(\frac{x \geq 1}{Y \leq 1}) = \frac{P(x \geq 2 \cap Y \leq 1)}{P(Y \leq 1)} = \frac{4K}{15K} = \frac{7}{15}$$
$$P(Y=2) = 24K + \frac{24}{72} = \frac{1}{3}$$
$$P(x=1) = 24K + \frac{1}{3}$$
$$P(x+y>3) = P(x=1, y=3) + P(x=2, y=2) + P(x=2, y=1)$$

Q. Joint PDF of R.V. X and Y is given as:

$$f_{xy}(x,y) = \begin{cases} 3xy; & x > 0; y > 0; \quad x + y \le 2 \\ 0; & 0 \neq 0 \le 2 \end{cases}$$



$$P[x + y \ge 1] = ?$$

4

$$\iint_R f_{xy}(x,y)\,dx\,dy = 1$$

$$P[x+y \ge 1] = \iint_{R^1} f_{xy}(x,y)\,dx\,dy$$

$$x + y \ge 1$$

$$x \to 1 - y \text{ to } 2 - y$$

$$x \to 0 \text{ to } 2 - y$$

$$\iint_R f_{xy}(x,y)\,dx\,dy = \iint_{R^1} f_{xy}(x,y) \cdot dx\,dy + \iint_{R''} f_{xy}(x,y)\,dx\,dy$$

$$1 - \iint_{R''} f_{xy}(x,y)\,dx\,dy = \iint_{R''} f_{xy}(x,y)\,dx\,dy$$

$$1 - \int_{x=0}^{1} \int_{y=0}^{y=1-x} 3/2\,xy\,dx\,dy = P[x+y \ge 1]$$

$$1 - 3/2 \times \frac{1}{4y} = 1 - \frac{1}{16} = \boxed{\frac{15}{16}}$$

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Statistical parameter of Random Variables:-

- Mean
- Mean Square value
- Variance
- Standard deviation
- nth order (Joint) moment about origin
- nth order (Joint) moment about mean
- Correlation
- Co-variance

- Correlation co-efficient
  1-0 R.V.
  2-0 R.V.

- Expectation operator:- E[x] = ?

(i) If PMF is given.

Expectation of RV x is given by $E[x] = \sum_i x_i P(x=x_i)$

Q. A random variable X represents the number of chocolates a child gets when spinning a wheel.

The possible outcomes and their probabilities are:

$P(X = 1) = 0.2$

$P(X = 2) = 0.5$

$P(X = 3) = 0.3$

Find the expected number of chocolates the child receives, i.e., E[X].

$E[x] = 1 \cdot P(x=1) + 2 \cdot P(x=2) + 3 \cdot P(x=3)$

$E[x] = 1(0.1) + 2(0.5)$

- 3(0.3)

= 0.2 1 1 + 0.9

$E[x] = 2.1$

$x \in 1, 2, 3$

$P(x) \in 0.2\ 0.5\ 0.3$

(ii) If PDF is given.

Expectation of R.V. x is given by $E[x] = \int_{-\infty}^{\infty} x f_X(x) \cdot dx$

Let X be a continuous random variable with the following PDF:

$f_X(x) = \{2x$ for $x \in [0, 1]$
0 otherwise

Find the expected value E[X].

$E[x] = \int_{-\infty}^{\infty} x f_Y(x) \cdot dx = \int_0^1 x\ x\ 2x \cdot dx = [2x^3/3]_0^1 = 2/3\ dx$

$E[x^2+3] = \int_{-\infty}^{\infty} (x^2+3) f_X(x) \cdot dx$ ✓

A Random variable y is defined as:

y = ax + b ; a and b are constants

Find E[y] = ? x is a R.V. whose pdf is $f_X(x)$.

E[y] = ∫₋∞^∞ y $f_Y(y)$ · dy
? ?

E[y] = E(ax+b) = ∫₋∞^∞ (ax+b) $f_X(x)$ · dx = ∫₋∞^∞ ax · $f_Y(x)$ · dx + ∫₋∞^∞ b $f_Y(x)$ · dx

= a E[x] + b

E(ax+b) = E(ax) + E[b] = a E[x] + b

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- Properties of Expectation operator:-

Let $x, y, g(x), x_1, x_2, x_3 \ldots$ are R.V.; a, b, c, τ are constants

(i) $E[ax \pm by] = E[ax] \pm E[by] = aE[x] \pm bE[y]$

(ii) $E[c] = c$

(iii) $E[ax_1 \pm bx_2 \pm cx_3 + \cdots \pm \alpha x_n] = aE[x_1] \pm bE[x_2] \pm cE[x_3] \pm \cdots \pm \alpha E[x_n]$

(iv) $E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x)\, dx$; PDF given

$= \sum_i g(x_i) \rho(x = x_i)$; PMF given

(vi) if $x \geq 0$; then $E[x] = \int_{-\infty}^{\infty} x f_x(x)\, dx \geq 0$

(vii) $E[(ax+b)^2] = E[a^2x^2 + b^2 + 2abx]$

$= a^2E[x^2] + b^2 + 2abE[x]$

$= a^2E[x^2] + 2abE[x] + b^2$

(viii) $E[xY] \neq E[x] \cdot E[Y]$

$\neq YE[x]$

$\neq xE[y]$

- Mean of a Random Variable:-

Mean of a R.V. = Expectation of a R.V.

(i) If PMF is given:-

$\mu_x = E[x] = \sum_i x_i \rho(x = x_i)$

(ii) If PDF is given:-

$\mu_x = E[x] = \int_{-\infty}^{\infty} x f_x(x)\, dx$

Q. A R.V. x is given. Another R.V. y is defined as $y = \sin x$

Find the mean of Random Variable y. PDF of R.V. x is given

as $f_x(x)$.

as $f_x(x)$.

- Moment about origin:-

1. 1^{st} order moment of R.V. x about origin:-

$E[(x-0)^2] = E[x] = \int_{-\infty}^{\infty} x f_x(x)\, dx \rightarrow \text{Mean of R.V. } x = \mu_x$

2. 2^{nd} order moment of R.V. x about origin:-

$E[(x-0)^2] = E[x^2] = \int_{-\infty}^{\infty} x^2 f_x(x)\, dx \rightarrow \text{Mean square value of R.V. } x$

3. n^{th} order moment of R.V. x about origin:-

$E[(x-0)^n] = E[x^n] = \int_{-\infty}^{\infty} x^n f_x(x)\, dx$

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- Moment about mean:-

$$E[x] = \mu_x$$

1. 1st order moment of R.V. x about mean:-

$$E[(x-\mu_x)^2] = E[x-\mu_x] = E[x] - E[\mu_x] = \mu_x - \mu_x = 0$$

2. 2nd order moment of R.V. x about mean:-

$$\begin{aligned} &E[(x-\mu_x)^2] = E[x^2 + \mu_x^2 - 2x\mu_x] = E[x^2] + E[\mu_x^2] - 2\mu_x E[x] \\ &= E[x^2] + \mu_x^2 - 2\mu_x^2 \\ &= E[x^2] - \mu_x^2 = E[x^2] - \{E[x]\}^2 \end{aligned}$$

$$\underbrace{E[(x-\mu_x)^2]}_{\downarrow \text{variance of R.V. } x} = E[x^2] - \{E[x]\}^2 = \sigma_x^2$$

- Conclusion:-

① Mean: - E[x] = ∫{-∞}^{∞} x f_y(x) dx
② Mean square value: - E[x^2] = ∫{-∞}^{∞} x^2 f_x(x) dx
③ Variance: - σ_x^2 = E[x^2] - {E[x]}^2
④ Standard deviation: - S.D. = √variance = √σ_x^2

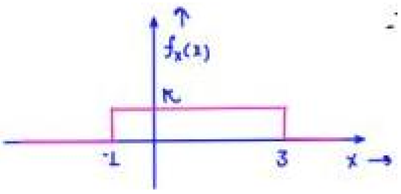
- Relationship w.r.t. the power:-

$$\sigma_x^2 = E[x^2] - \{E[x]\}^2 \Rightarrow \begin{array}{c} \int_{-\infty}^{\infty} d\tau \text{ power of R.V. } x \\ E[x^2] = \sigma_x^2 + \{E[x]\}^2 \\ \downarrow \\ \text{MSV of R.V. } x \\ \text{Total power of R.V. } x \end{array} \quad \downarrow \text{DC power of R.V. } x$$

Q. The POF of R.V. is shown.

Find it's mean and variance.

$$\int_{-\infty}^{\infty} f_x(x)\,dx = 1$$



$$K \times 4 = 1$$

$$K = 1/4$$

$$\text{Mean: - } E[x] = \int_{-\infty}^{\infty} x f_y(x)\,dx = \int_{-1}^{3} \frac{x}{4}\,dx = \frac{1}{4}\left[\frac{x^2}{2}\right]_{-1}^{3}$$

$$f_y(x) = \begin{cases} 1/4; & -1 < x < 3 \\ 0; & 0 \le x \le 3 \end{cases}$$

$$= \frac{1}{8}[9 \cdot 1] = 1$$

$$E[x] = \mu_x = \bar{x} = 1$$

$$\sigma_x^2 = E[x^2] - \{E[x]\}^2$$

$$E[x^2] = \int_{-\infty}^{\infty} x^2 f_x(x)\,dx = \int_{-1}^{3} \frac{x^2}{4}\,dx = \frac{1}{12}[x^3]_{-1}^{3}$$

$$= \frac{1}{12}[21 + 1] = \frac{7}{3}$$

$$\sigma_x^2 = \frac{7}{3} - (1)^2 = 4/3$$

$$\mu_x = 1$$

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Q. A R.V. Y is defined as $Y = ax + b$

Find mean and variance of Y.

$Y = ax + b$

$E[Y] = E(ax+b) = aE[x] + b$

$\sigma_Y^2 = E[Y^2] - \{E[Y]\}^2$
4 $E[Y^2] = E[(ax+b)^2] = E[a^2x^2 + b^2 + 2abx] = a^2E[x^2] + b^2 + 2abE[x]$
$\rightarrow \sigma_Y^2 = a^2E[x^2] + b^2 + 2abE[x] - \{aE[x] + b\}^2$
$= a^2E[x^2] + b^2 + 2abE[x] - a^2\{E[x]\}^2 - b^2 - 2abE[x]$

$\sigma_Y^2 = a^2E[x^2] - a^2\{E[x]\}^2$

$= a^2[E(x^2) - \{E(x)\}^2]$

$\sigma_Y^2 = a^2\sigma_Y^2$

$Y = ax + b$

$\mu_Y = a\mu_Y + b$

$\sigma_Y^2 = a^2\sigma_Y^2$

$E[Y^2] = \sigma_Y^2 + \mu_Y^2$

Conclusion:

$Y = ax + b$

$\mu_Y = a\mu_Y + b$

$\sigma_Y^2 = a^2\sigma_Y^2$

Mean = dry = DC value

= Expectation

$= E[x] = \mu_x$

Q.

$Y = -2x + 3$; $\mu_x = 3$; $\sigma_x^2 = 4$

$\sigma_Y^2 = ?$, $\mu_Y = ?$;

; $\sigma_Y^2 = 16$ $\mu_Y = -3$

Total power of R.V. X / Total power of R.V. Y = ? = 18 / 25

$\sigma_Y^2 = (-1)^2\sigma_Y^2$

$= 4 \times 4 = 16$

$\mu_Y = -2\mu_Y + 3$

$= -2(3) + 3 = -3$

$E[x^2] = \sigma_x^2 + \mu_x^2 = 4 + 9 = 13$

$E[Y^2] = 16 + 9 = 25$

Q. A R.V. X takes values from 1 to 5 with probabilities

as shown. Find mean and variance.

x	1	2	3	4	5
P(x=x)	0.1	0.2	0.4	0.2	0.1

$P[x^2 = u] = P[x=2] + P[x=4] = 0.2$

$= 0.2$

4

$E[x] = 0.1 + 0.4 + 1.2 + 0.8 + 0.5 =$

$\sigma_x^2 = E[x^2] - \{E[x]\}^2$

$3 = \mu_x$

$E[x^2] = \Sigma_i x_i^2 P(x=x_i)$

$x^2 \rightarrow 1$	4	9	16	25
$P \rightarrow 0.1$	0.2	0.4	0.2	0.1

$E[x^2] = 0.1 + 0.3 + 3.6 + 3.2 + 2.5$

$E[x^2] = 10.2$

$\sigma_x^2 = 10.2 - 9 = 1.2$

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Q. PDF of a R.V. is given $f_x(z) = \frac{1}{2}|x|e^{-|x|}$; $-\infty < z < \infty$

Find mean, variance and 11^{th} order moment about
= 0 = 6 = 0 origin.

4 (a) $E[x] = \int_{-\infty}^{\infty} x f_x(z)\,dz = \frac{1}{2}\int_{-\infty}^{\infty} x|x|e^{-|x|}\,dz$
$= \frac{1}{2}\int_{-\infty}^{0} x^2 e^{x}\,dx + \frac{1}{2}\int_{0}^{\infty} x^2 e^{-x}\,dx$
$\int_{-\infty}^{\infty} odd \text{ function } dx = 0$
$\quad x = -t$
$\quad dx = -\infty$
$-\frac{1}{2}\int_{0}^{\infty} t^2 e^{-t}\,dt$
$E[X^{(1)}] = \frac{1}{2}\int_{-\infty}^{\infty} x^{(1)}|x|e^{-|x|}\,dx$
$E[X^{(2)}] = 0$

(3) $\sigma_x^2 = E[x^2] - \{E[x]\}^2$
$E[x^2] = \int_{-\infty}^{\infty} x^2 f_x(z)\,dz = \frac{1}{2}\int_{-\infty}^{\infty} x^2|x|e^{-|x|}\,dz$
$\sigma_x^2 = 6 - (0)^2 = 6$
even

$e^{-t}u(t) \longrightarrow \frac{L}{s+L}$
$= 2 \times \frac{1}{2}\int_{0}^{\infty} x^2|x|e^{-|x|}\,dx = \int_{0}^{\infty} x^3 e^{-x}\,dx$
$= 6 \quad \text{plus}$
$te^{-t}u(t) \longrightarrow \frac{L}{(s+L)^2}$
$t^2e^{-t}u(t) \longrightarrow \frac{2}{(s+L)^3}$
$t^3e^{-t}u(t) \longrightarrow \frac{6}{(s+L)^4}$
$x(s) = \int_{-\infty}^{\infty} x(t)\cdot e^{-st}\,dt$
$x(s)|_{s=0} = \int_{-\infty}^{\infty} x(t)\,dt = \int_{-\infty}^{\infty} t^3 e^{-t}u(t)\,dt = \int_{0}^{\infty} t^3 e^{-t} dt$
$\frac{6}{(s+L)^4}|_{s=0} = 6$

Q. Let the R.V. x represents the no. of times dice needs to be thrown till 3 is observed for the first time.
Find expectation of x.

4. x: no. of times dice needs to be thrown till 3 is observed.

$S = \{3, \bar{3}3, \bar{3}\bar{3}3, \bar{3}\bar{3}\bar{3}3 \ldots\ldots\ldots\}$

$x \in \{1, 2, 3, 4 \ldots\ldots\ldots\}$

$P(x=1) = \frac{1}{6}$; $P(x=2) = \frac{5}{6} \times \frac{1}{6}$; $P(x=3) = \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6}$; $\ldots\ldots$

$E[x] = 1 \times \frac{1}{6} + 2 \times \frac{5}{6} \times \frac{1}{6} + 3 \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + 4 \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \ldots\ldots$

$E[x] = 1 \times \frac{1}{6} + 2 \times \frac{5}{6} \times \frac{1}{6} + 3 \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + 4 \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \ldots\ldots$

$\frac{5}{6}E[x] = \frac{5}{6} \times \frac{1}{6} + 2 \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + 3 \times \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \ldots\ldots$

$\frac{E[x]}{6} = 1 \times \frac{1}{6} + \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \ldots\ldots$

$\frac{E[x]}{6} = \frac{1}{1-\frac{5}{6}} = 1$
$E[x] = 6$

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- Joint moment about origin :-

$$E[x] = \int_{-\infty}^{\infty} x f_x(x)\, dx$$

① Joint moment of order (1,1) about the origin for RV x a y.

$$E[(x-0)(y-0)]^2 = E[xy] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy \cdot f_{xy}(x,y) \cdot dx\, dy$$

Correlation Mw = E[xy] = Rxy =

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf_{xy}(x,y)\, dx\, dy$$

R.V. x and Y

② Joint moment of order (r, k) about the origin for RV x and y.

$$E[x^r y^k] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^r y^k f_{xy}(x,y)\, dx\, dy$$

- Joint moment about mean :-

$$E[x] = \mu_x$$

$$E[y] = \mu_y$$

Joint moment of order (1,1) about the mean for RV x a y.

$$E[(x-\mu_x)^2(y-\mu_y)^2] = E[(x-\mu_x)(y-\mu_y)] = E[xy - x\mu_y - \mu_x y + \mu_x\mu_y]$$

$$= E[xy] - \mu_y E[x] - \mu_x E[y] + \mu_y\mu_y$$

$$= E[xy] - \mu_y\mu_x - \mu_y\mu_y + \mu_x\mu_y$$

$$E[(x-\mu_x)(y-\mu_y)] = E[xy] - E[x]\cdot E[y]$$

Covariance of R.V. x and y : Cxy = σxy = E[xy] - E[x]·E[y] = Rxy - μxμy

- Conclusion :-

① Correlation between R.V. x and y :- E[xy] =

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf_{xy}(x,y)\,dx\,dy$$

Covariance between R.V. x and y :- σxy = Cxy = E[xy] - E[x]·E[y]

② Correlation co-efficient

$$f_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

Cxy = Rxy - μxμy

R.V. x

$$\longrightarrow$$

μx, σx², fx(x)

R.V. y

$$\longrightarrow$$

μy, σy², fy(y)

R.V. x, y

$$\longrightarrow$$

fxy(x,y), Rxy, σxy or Cxy

Q. w = âx̂ + b̂ŷ

Find mean, MSV and variance of w.

E[w] = E(ax+by) = aE[x]+bE[y] = aμx+bμy ✓

$$E[w^2] = E[a^2x^2 + b^2y^2 + 2abxy] = a^2E[x^2] + b^2E[y^2] + 2abE[xy]$$

✓

$$\sigma_w^2 = E[w^2] - \{E[w]\}^2$$

$$= a^2E[x^2] + b^2E[y^2] + 2abE[xy] - a^2\mu_x^2 - b^2\mu_y^2 - 2ab\mu_x\mu_y$$

$$= a^2\{E[x^2] - \mu_x^2\} + b^2\{E[y^2] - \mu_y^2\} + 2ab\{E[xy] - E[x]\cdot E[y]\}$$

$$\sigma_w^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\sigma_{xy}$$

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Conclusion:- $$z = ax + b$$ $$\sigma_z^2 = a^2\sigma_x^2$$ $$w = ax + by$$ $$\text{var}(w) = \sigma_w^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\sigma_{xy}$$	$$\rho_{\text{covariance}}$$ $$\sigma_{xy} = E[xy] - E[x]E[y]$$
$$\rho E[x]$$ $$\bar{x} = 1/2$$, $$\rho E[x^2]$$ $$\bar{y} = 2$$, $$\bar{y}^2 = 1\frac{1}{2}$$, $$C_{xy} = -\frac{1}{2\sqrt{3}} = \sigma_{xy}$$	$$\rho_{\text{covariance}}$$ $$\sigma_x^2 = 5/2 - 1/4 = \frac{9}{4}$$
$$w = 3x - 2y$$ i) find mean, Msv and variance of w ii) find correlation co-efficient of Rv x and y.	$$\sigma_y^2 = \frac{19}{2} - 4 = \frac{11}{2}$$
$$E[w] = 3E[x] - 2E[y] = 8/2 - 2 \times 2 = -5/2$$ $$E[w^2] = E[3x^2 + 4y^2 - 12xy] = 9E[x^2] + 4E[y^2] - 12E[xy]$$	
$$C_{xy} = E[xy] - E[x] \cdot E[y]$$	

$$-\frac{1}{2\sqrt{3}} = E[xy] - 1$$ $$\Rightarrow$$ $$E[xy] = 1 - \frac{1}{2\sqrt{3}}$$	
$$E[w^2] = \frac{9\times 5}{2} + \frac{4\times 19}{2} - 12\left[1 - \frac{1}{2\sqrt{3}}\right]$$ $$= 45/2 + 38 - 12 + 6/\sqrt{3} = 51.96$$ $$w = 3x - 2y$$ $$\sigma_w^2 = 9\sigma_x^2 + 4\sigma_y^2 - 12C_{xy}$$	
$$4\sigma_w^2 = 9\left(\frac{9}{4}\right) + 4\left(\frac{11}{2}\right) - 12\left[-\frac{1}{2\sqrt{3}}\right] = 45.71$$	
$$4\sigma_w^2 = E[w^2] - \{E[w]\}^2 = 51.96 - \{-5/2\}^2 = 51.96 - \frac{25}{4} = 45.71$$	
$$4f_{xy} = \frac{\sigma_{xy}}{\sigma_x\sigma_y} = \frac{-1/2\sqrt{3}}{3/2 + \sqrt{11/2}}$$; $$\sigma_x^2 = 9/4$$ $$\sigma_y = 9/2$$ $$\sigma_y^2 = 11/2$$	

;

$$\sigma_y = \sqrt{11/2}$$

Important NOTE:-

$$E[(2x+3)(y^2+1)] = E[1x+3) \cdot E[y^2+1]$$

Random Variable x and y are Orthogonal

$$E[xy] = 0$$

$$R_{xy} = 0$$

$$C_{xy} = E[xy] - E[x] \cdot E[y]$$

$$C_{xy} = -\mu_x \mu_y$$

Uncorrelated

$$E[xy] = E[x] \cdot E[y]$$

$$R_{xy} = \mu_x \mu_y$$

$$C_{xy} = E[xy] - E[x] \cdot E[y]$$

$$C_{xy} = 0 = \sigma_{xy}$$

Independent

$$E[g(x) \cdot h(y)] = E[g(x)] \cdot E[h(y)]$$

$$f_{xy}(x,y) = f_x(x) \cdot f_y(y)$$

N.B.- ① If x and y are independent then :-

$$E[g(x) \cdot h(y)] = E[g(x)] \cdot E[h(y)]$$

$$E_g \cdot E[(x^2+3)(y^2+2y+4)] = E[x^2+3] \cdot E[y^2 -$$

② If x and y are uncorrelated then :-

$$E[xy] = E[x] \cdot E[y]$$

$$E[g(x) \cdot h(y)] \neq E[g(x)] \cdot E[h(y)]$$

③ If x and y are independent then they are s uncorrelated. If x and y are uncorrelated the may or may not be independent.

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Q.

$$f_{xy}(x,y) = \begin{cases} \frac{1y}{3}; & 0 < x < 2 \text{ and } 0 < y < 3 \\ 0; & 0/\infty \end{cases}$$

Find covariance of x and y.

check independency, orthogonality and un-correlation.

$$C_{xy} = E[xy] \cdot E[x] \cdot E[y]$$

$$E[xy] = \iint_R xyf_{xy}(x,y) \cdot dxdy$$

$$E[x] = \int_R^R x \cdot f_y(x) \cdot dx \quad ; \quad E[y] = \int_y y \cdot f_y(y) \cdot dy$$

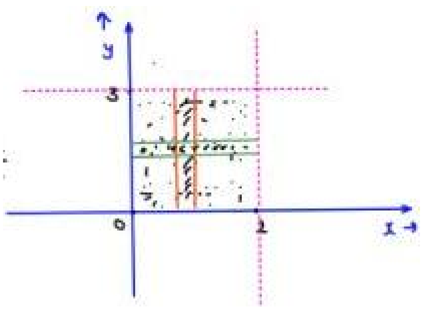
$$f_x(x) = \int_{y=0}^{3} f_{xy}(x,y)dy = \int_{y=0}^{3} \frac{xy}{9}dy$$

$$= \frac{x}{9}\left[\frac{y^2}{2}\right]_0^3 = \frac{x}{2}$$

$$f_x(x) = x/2 \quad ; \quad 0 < x < 2$$

$$f_y(y) = \int_{x=0}^{2} \frac{xy}{9}dx = \frac{y}{9}\left[\frac{x^2}{2}\right]_0^2 = \frac{2y}{9}$$

$$f_y(y) = \frac{2y}{9} \quad ; \quad 0 < y < 3$$



$$f_{xy}(x,y) = f_y(x) \cdot f_y(y)$$

4 x and y are independent.

4 x and y are uncorrelated.

$$E[xy] = \int_{y=0}^{3}\int_{x=0}^{2} xyf_{xy}(x,y) \cdot dxdy = \int_{y=0}^{3}\int_{x=0}^{2} xy \cdot \frac{xy}{9} \cdot dxdy$$

$$= \frac{1}{9}\left[\frac{x^3}{3}\right]_0^2\left[\frac{y^3}{3}\right]_0^3 = \frac{1}{9} \times \frac{8}{3} \times 9$$

$$E[x] = \int_0^2 xf_y(x)dx = \int_0^2 \frac{x^2}{2}dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{8}{6} = 8/3$$

$$E[y] = \int_{y=0}^{3} y f_y(y) dx = \int_0^3 \frac{2y^2}{9} dy = \left[\frac{2y^3}{27}\right]_0^3 = 2$$

$$E[xy] = 8/3 \neq 0 \quad ; \quad E[x] = 8/6 \quad ; \quad E[y] = 2$$

$$E[xy] = E[x] \cdot E[y]$$

Q.

$$z = -8x + 4y$$

Find variance of R.V. z when

(a) R.V x and y are uncorrelated
(b) R.V x and y are orthogonal.
(c) R.V x and y are independent.

4. $$\sigma_z^2 = 9\sigma_x^2 + 16\sigma_y^2 - 24c_{xy}$$

(c) (a) x and y are uncorrelated.

$$E[xy] = E[x] \cdot E[y]$$

;

$$c_{xy} = E[xy] \cdot E[x] \cdot E[y]$$

$$\boxed{\sigma_z^2 = 9\sigma_x^2 + 16\sigma_y^2}$$

(b) x and y are orthogonal:

$$E[xy] = 0$$

;

$$c_{xy} = -\mu_x \mu_y$$

$$\boxed{\sigma_z^2 = 9\sigma_x^2 + 16\sigma_y^2 + 24\mu_x\mu_y}$$

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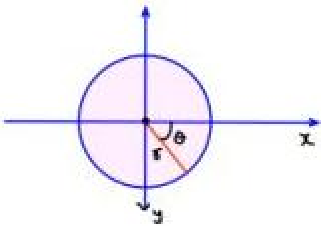
ORV
$x \in \{-s, 0, 1\}; \quad P(x=-1) = P(x=0) = P(x=1) = 1/3$
$\sum_{y=x^2}^{P(x)} \rightarrow x$ and y are depending on each other.
Find $E[x], E[y], E[xy] = ?$
Q. Comment on independency, orthogonality and uncorrelation?
$E[x] = (-1)(1/3) + (0)(1/3) + (1)(1/3) = 0$
$E[y] = E[x^2] = (-1)^2(1/3) + (0)^2(1/3) + (1)^2(1/3) = 2/3$
$E[xy] = E[x^3] = (-1)^3(1/3) + (0)^3(1/3) + (1)^3(1/3) = 0$
$\underbrace{E[xy] = E[x] \cdot E[y]}_{\text{uncorrelation}} \quad \underbrace{E[yy] = 0}_{\text{orthogonality}}$

Q. If $f_x(x) = f_x(-x) \rightarrow$ POF of R.v. x is even.
Find $E[x] = ?$
$E[x] = 0$
$g(x) = x$
$g(-x) = -x = -g(x)$
$E[x] = 0$
odd
$h(x) = x f_x(x)$ h(-x) = -x f_x(-x) f_x(x) = f_x(-x)h(-x) = -x f_x(x)h(-x) = -h(x)$

Q. $f_{xy}(1, y) = \begin{cases} \frac{6}{\pi}(x^2 + y^2); & x^2 + y^2 \leq 1 \\ 0; & 0/\omega \end{cases}$
$\underbrace{x^2 + y^2 = 1}_{\text{circle}}$
comment on orthogonality, independency and un-correlation?
$E[xy] = \iint_R xy f_{xy}(x, y)\, dx\, dy$
or x and y will be taking
values such that $x^2 + y^2 \leq 1$
x and y are depending on each other
$\downarrow$
not independent.



$x = r\cos\theta$

$y = r\sin\theta$

$dx\, dy = r\, dr\, d\theta$

$\theta \in [0, 2\pi]$

$r \in [0,1]$

$E[xy] = \iint_R xy f_{xy}(x,y)\,dx\,dy = \iint_R xy \times \frac{6}{\pi}(x^2+y^2)\,dx\,dy$

$E[xy] = 0 \to x$ and y
are orthogonal

$$= \iint_{\theta=0}^{2\pi} r^2 \cos\theta \sin\theta \times \frac{6}{\pi} r^2 \cdot r\,dr\,d\theta$$
$$= \frac{6}{\pi} \int_{r=0}^{2\pi} r^5 \cdot dr \int_{\theta=0}^{2\pi} \frac{\sin 2\theta}{\alpha} d\theta = 0$$

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$$E[x] = \int x f_x(x) \cdot dx \quad f_x(x) = ?$$
$$f_x(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} f_{xy}(x,y)\,dy = \frac{6}{\pi} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} (x^2+y^2) \cdot dy$$
$$-1 < x < 1; \quad f_x(x) = f_x(-x)$$
$$\boxed{E[x]=0} \quad = \frac{6}{\pi} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x^2 \cdot dy + \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} y^2 \cdot dy$$
$$= \frac{6}{\pi} x^2 [2\sqrt{1-x^2}] + \left[\frac{y^3}{3}\right]_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}$$

$$f_x(x) = \frac{6}{\pi} x^2 [2\sqrt{1-x^2}] + \frac{2}{3}(1-x^2)^{\frac{3}{2}}; \quad -1 < x < 1$$

$$f_y(y) = \frac{6}{\pi} y^2 [2\sqrt{1-y^2}] + \frac{2}{3}(1-y^2)^{\frac{3}{2}}; \quad -1 < y < 1$$

$$\Downarrow$$

$$-1 < y < 1; \quad f_y(y) = f_y(-y)$$

$$E[y] = 0$$

$$E[xy] = 0 = E[x] = E[y] \quad \checkmark \quad \text{orthogonal} \checkmark$$
$$E[xy] = E[x] \cdot E[y] \quad \checkmark \quad \text{uncorrelated} \checkmark$$

$$f_{xy}(x,y) \neq f_x(x) \cdot f_y(y); \quad \text{not independent}$$

Q.

$$f_{xy}(x,y) = \begin{cases} 6(1-y); & 0 \le x \le y \le 1 \\ 0; & 0/\omega \end{cases}$$

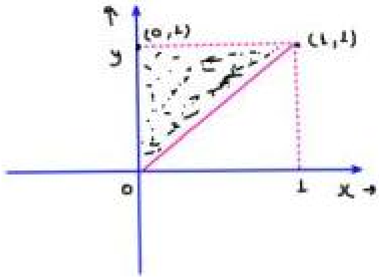
$$x \le y$$

x and y are independent?

4

$$\Downarrow$$

$$\text{No}$$



Q. x and y are two R.V.; $\bar{x} = 1, \bar{y} = 2, \sigma_x^2 = 6, \sigma_y^2 = y$

$$4 \; f_{xy} = -2/3$$

Covariance and correlation = ?

$$\sigma_x = \sqrt{6}$$

5

$$f_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

$$\sigma_y = 3$$

$$\sigma_{xy} = -2/3 \times \sqrt{6} \times 3 = -2\sqrt{6}$$

$$R_{xy} = E[xy] = ?$$

$$R_{xy} = -2\sqrt{6} + 2 \quad \checkmark$$

$$\text{dim}$$

$$\sigma_{xy} = E[xy] - E[x] \cdot E[y]$$

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Q. Correct statement?

(a) If x and y are orthogonal then $\sigma_{xy} = -\mu_x \mu_y$

(b) If x and y are uncorrelated then $R_{xy} = \mu_x \mu_y$

(c) If x and y are orthogonal then $R_{xy} = 0$

(d) If x and y are uncorrelated then $\sigma_{xy} = 0$

orthogonal; $E[xy] = 0$; $\sigma_{xy} = 0 - \mu_x \mu_y = -\mu_x \mu_y$
$R_{xy} = 0$

uncorrelation; $E[xy] = E[x] \cdot E[y]$; $\sigma_{xy} = R_{xy} - \mu_x \mu_y$
$R_{xy} = \mu_x \mu_y = 0$

- Some Mathematics and important points:

$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} = 2\int_{0}^{\infty} e^{-x^2} \cdot dx$

$\int_{0}^{\infty} e^{-x^2} \cdot dx = \frac{\sqrt{\pi}}{2}$

$\int_{-\infty}^{\infty} e^{-x/2} \cdot dx = \sqrt{2\pi}$

$= \int_{-\infty}^{\infty} e^{-(x/\sqrt{2})^2} \cdot dx = \sqrt{\pi} \times \sqrt{2} = \sqrt{2\pi}$

$e^{-ax^2} \to \sqrt{\pi a} e^{-a^2/4a}$

$\int_{-\infty}^{\infty} f(t) \cdot dt = A$

$\int_{-\infty}^{\infty} f(t/a) \cdot dt = aA$

$t/a = \alpha$

$dt = ad\alpha$

$a\int_{-\infty}^{\infty} f(\alpha) \cdot d\alpha = aA$

4. Q-function:

$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} e^{-t^{1/2}} \cdot dt$

4. $Q(-\infty) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^{1/2}} \cdot dt = \frac{1}{\sqrt{2\pi}} \times \sqrt{2\pi} = L$
   $Q(0) = \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-t^{1/2}} \cdot dt = \frac{1}{\sqrt{2\pi}} \times \sqrt{\frac{2\pi}{2}} = L/2$
   $Q(\infty) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^{1/2}} \cdot dt = 0$

- $\boxed{Q(x) + Q(-x) = L}$
  NOTE: (i) $Q(-x) = 1 - Q(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^{1/2}} \cdot dt$

(ii) complementary error function $erfc(x) = 2Q(\sqrt{2}x)$

$= \frac{2}{\sqrt{\pi}} \int_{x}^{\infty} e^{-t^2} \cdot dt$
(iii) $Q(x) = \frac{1}{2} erfc\left(\frac{x}{\sqrt{2}}\right) = \frac{1}{2}\left[1 - erf\left(\frac{x}{\sqrt{2}}\right)\right]$
(iv) Error function $erf(x) = L - erfc(x)$
$= \frac{2}{\pi} \int_{0}^{x} e^{-t^2} \cdot dt$

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Q. You did a certain calculation and your answer came out to be Q(0.4) but all the options are mentioned

in terms of p(x).

where

$$P(x) = \int_0^x e^{-t^2} \cdot dt$$

write down Q(0.4) in terms of p(x)?

4. (a)

$$\frac{1}{2} - \frac{1}{\sqrt{\pi}} P\left(\frac{0.4}{\sqrt{2}}\right)$$

(c)

$$\frac{1}{2} - \frac{1}{\sqrt{\pi}} P(0.2)$$

(b)

$$\frac{1}{2} + \frac{1}{\sqrt{\pi}} P\left(\frac{0.4}{\sqrt{2}}\right)$$

(d)

$$\frac{1}{2} + \frac{1}{\sqrt{\pi}} P(0.2)$$

4. Q(0.4) √

$$P(x) = \int_0^x e^{-t^2} \cdot dt$$

$$P\left(\frac{x}{\sqrt{2}}\right) = \int_0^{\frac{x}{\sqrt{2}}} e^{-t^2} \cdot dt$$

$$\int_0^{\infty} e^{-t^2} \cdot dt = \frac{\sqrt{\pi}}{2}$$

4.

$$\int_0^{\frac{x}{\sqrt{2}}} e^{-t^2} \cdot dt + \int_{\frac{x}{\sqrt{2}}}^{\infty} e^{-t^2} \cdot dt = \frac{\sqrt{\pi}}{2}$$

$$P\left(\frac{x}{\sqrt{2}}\right) + \sqrt{\pi} \cdot Q(x) = \frac{\sqrt{\pi}}{2}$$

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_2^{\infty} e^{-t^2/2} \cdot dt$$

$$t_{\sqrt{2}} = \alpha \Rightarrow dt = \sqrt{\alpha}\, d\alpha$$

$$= \frac{\sqrt{\alpha}}{\sqrt{2\pi}} \int_{\frac{x}{\sqrt{2}}}^{\infty} e^{-\alpha^2} \cdot d\alpha$$

$$\sqrt{\pi} \cdot Q(x) = \int_{\frac{x}{\sqrt{2}}}^{\infty} e^{-t^2} \cdot dt$$

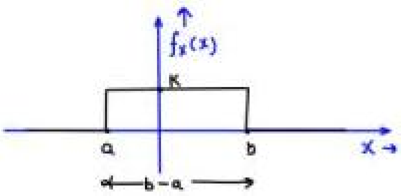
$$Q(x) = \frac{1}{2} - \frac{1}{\sqrt{\pi}} P\left(\frac{x}{\sqrt{2}}\right)$$

$$Q(0.4) = \frac{1}{2} - \frac{1}{\sqrt{\pi}} P\left(\frac{0.4}{\sqrt{2}}\right)$$

- Some Standard distribution:

1. Uniform distribution: cev

$$x \sim U[a, b]$$



$$\int_{-\infty}^{\infty} f_p(x)\, dx = 1$$

$$k(b-a) = 1 \Rightarrow k = \frac{1}{b-a}$$

$$f_p(1) = \begin{cases} \frac{1}{b-a} & ; \quad a < x < b \\ 0 & ; \quad 0 < w \end{cases}$$

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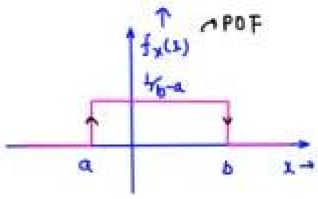
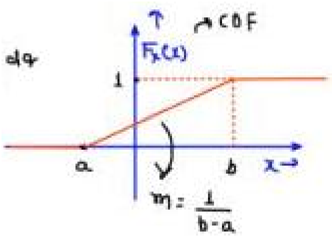


$$F_x(x) = \int_{-\infty}^{x} f_\theta(s)\, ds$$



① Mean:-

$$E[x] = \int_{-\infty}^{\infty} x f_x(x)\, dx = \int_a^b x \cdot \frac{1}{b-a}\, dx = \frac{1}{b-a}\left[\frac{x^2}{a}\right]_a^b$$

$$= \frac{1}{2(b-a)}(b-a)(b+a)$$

$$E[x] = \frac{a+b}{2}$$

② Mean Square value:

$$E[x^2] = \int_{-\infty}^{\infty} x^2 f_x(x)\, dx = \int_a^b \frac{x^2}{b-a}\, dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b$$

$$= \frac{1}{3} \times \frac{1}{b-a}[b-a][b^2+a^2+ab] \quad ; \quad E[x^2] = \frac{a^2+b^2+ab}{3}$$

③ Variance:

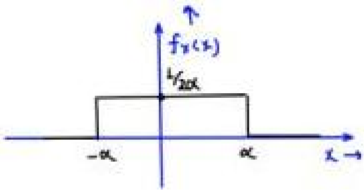
$$\sigma_x^2 = E[x^2] - \{E[x]\}^2 = \frac{a^2+b^2+ab}{3} - \frac{a^2+b^2+2ab}{4}$$

$$\sigma_x^2 = \frac{(b-a)^2}{12} \quad E[x] = \frac{a+b}{2} = \frac{4a^2+4b^2+4ab-3a^2-3b^2-6ab}{12}$$

$$= \frac{a^2+b^2-2ab}{12} = \frac{(b-a)^2}{12}$$

If the POF is symmetric about origin:-

$$x \sim \cup[-\alpha, \alpha]$$



Mean:-

$$E[x] = \frac{a+b}{2} = -\frac{\alpha+\alpha}{2} = 0$$

Variance:-

$$\sigma_x^2 = \frac{(b-a)^2}{12} = \frac{(2\alpha)^2}{12} = \frac{\alpha^2}{3}$$

$$E[x] = 0$$

$$\sigma_x^2 = \frac{\alpha^2}{3}$$

Q.

$$x \sim \cup[-1, 3]$$

; mean, MSV and variance = ?

$$E[x] = \frac{3-2}{2} = 0.5$$

$$\sigma_x^2 = \frac{(5)^2}{12} = \frac{25}{12}$$

$$E[x^2] = \sigma_x^2 + \{E[x]\}^2$$

$$= \frac{25}{12} + \frac{1}{4} = \frac{28}{12}$$

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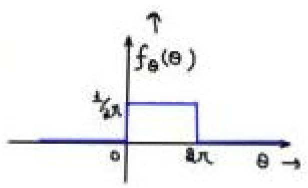
Q. a R.V. x is a f^n of another R.V. θ



R.V. → x = 3 cos θ

R.V. θ ~ U[0, 2π]

Find variance of R.V. x = ?

$\sigma_x^2 = E[x^2] - \{E[x]\}^2$

$E[x^2] = E[3\cos^2\theta] = g\, E[\cos^2\theta] = g \int_0^\infty \cos^2\theta \cdot f\theta(\theta) \cdot d\theta$
$= g \int_0^{2\pi} \cos^2\theta / 2\pi \cdot d\theta = g/2\pi \int_0^{2\pi} (1 + \cos 2\theta)/2 \cdot d\theta$
$= g/2\pi\, [2\pi/2] = 9/2$

$E[x] = E[3\cos\theta] = \int_{-\infty}^{\infty} 3\cos\theta \cdot f\theta(\theta) \cdot d\theta = \int_0^{2\pi} (3\cos\theta / 2\pi) \cdot d\theta$

$E[x] = 0$

$= 0$

$\sigma_x^2 = E[x^2] - \{E[x]\}^2$

$\sigma_x^2 = 9/2$

2. Gaussian Distribution / Normal Distribution:

X ~ N(2, 3)

E[x] = 2

$\sigma_x^2 = 3$

Mean = μ_x

Variance = σ_x^2

$f_x(x) = (1/\sqrt{(2\pi\sigma_x^2)})\ e\verb|^|(-(x - \mu_x)^2/2\sigma_x^2)$

4 $Q(x) = (1/\sqrt{(2\pi)}) \int_x\verb|^|\infty\ e\verb|^|(-x^2/2)\ dx$

$\int_{-\infty}\verb|^|\infty\ e\verb|^|(-x^2) \cdot dx = \sqrt{\pi}$

$\int_{-\infty}\verb|^|\infty\ f_x(x) \cdot dx = 1$

$\int_{-\infty}\verb|^|\infty\ e\verb|^|(-(x-2)^2) \cdot dx = \sqrt{\pi}$

$(1/\sqrt{(2\pi\sigma_x^2)}) \int_{-\infty}\verb|^|\infty\ e\verb|^|(-(x-\mu_x)^2/2\sigma_x^2 \cdot dx$

$= (1/\sqrt{(2\pi\sigma_x^2)})\ \sqrt{(1/\sqrt{(2\pi x)^2})} = 1$

① Mean:
$E[x] = \int_{-\infty}\verb|^|\infty\ x\ f_x(x) \cdot dx = \int_{-\infty}\verb|^|\infty\ x \cdot (1/\sqrt{(2\pi\sigma_x^2)})\ e\verb|^|(-(x - \mu_x)^2/2\sigma_x^2 \cdot dx$

$E[x] = \mu_x$

② Mean Square Value:
$E[x^2] = \int_{-\infty}\verb|^|\infty\ x^2\ f_x(x) \cdot dx = \int_{-\infty}\verb|^|\infty\ x^2 \cdot (1/\sqrt{(2\pi\sigma_x^2)})\ e\verb|^|(-(x - \mu_x)^2/2\sigma_x^2 \cdot dx = \sigma_x^2 + \mu_x^2$

③ Variance: Variance = σ_x^2

$\sigma_x^2 = E[x^2] - \{E[x]\}^2$
$E[x^2] = \sigma_x^2 + \mu_x^2$

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Q. we know that

$$\int_{-\infty}^{\infty} e^{-x^2} \cdot dx = \sqrt{x}$$

find

$$\int_{-\infty}^{\infty} e^{-(x-2)^2} \cdot dx = ? = \sqrt{\pi}$$

4

$$x - 2 = t$$

$$dx = dt$$

$$\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$$

Q.

$$I = \int_{-\infty}^{\infty} x^2 \exp\left[-\frac{(x-2)^2}{4}\right] \cdot dx$$

; find

$$I = ?$$

4

$$f_x(1) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{-(x-\mu_x)^2/2\sigma_x^2}$$

$$\mu_x = 2$$

$$\sigma_x^2 = 2$$

$$f_x(1) = \frac{1}{\sqrt{4\pi}} e^{-(x-2)^2/4}$$

$$\int_{-\infty}^{\infty} \frac{x^2}{\sqrt{4\pi}} e^{-(x-2)^2/4} \cdot dx = 6$$

$$E[x^2] = \sigma_x^2 + \mu_x^2 = 2 + 4 = 6$$
$$6 = \int_{-\infty}^{\infty} x^2 f_x(x) \cdot dx$$

$$\int_{-\infty}^{\infty} x^2 e^{-(x-2)^2/4} \cdot dx = \boxed{6\sqrt{4\pi} = 1}$$

$$I = 12\sqrt{\pi}$$

Q.

$$I = \int_0^{\infty} x \cdot e^{-x^2} \cdot dx$$

$$x^2 = t$$

$$2xdx = dt$$

$$I = \int_0^{\infty} \frac{e^{-t}}{2} dt = \frac{1}{2} \int_0^{\infty} e^{-t} \cdot dt = 1/2$$

$$I = 1/2$$

Q.

$$f_x(x) = k \cdot e^{-x^2/2}$$

; find

$$k = ?$$

$$\int_{-\infty}^{\infty} f_x(x) \cdot dx = 1$$

;

$$\int_{-\infty}^{\infty} ke^{-x^2/2} \cdot dx = 1$$

$$k = \sqrt{2\pi} = 1$$

$$4 f_x(x) = \frac{1}{\sqrt{2\pi}\sigma_x^2} e^{-(x-\mu_x)^2/2\sigma_x^2}$$

$$k = \frac{1}{\sqrt{2\pi}}$$

$$\mu_x = 0$$

$$\sigma_x^2 = 1$$

$$k = \frac{1}{\sqrt{2\pi}} V$$

Q.

$$I = \int_{2}^{\infty} e^{-(x-2)^2/30} \cdot dx$$

4

$$x - 2 = t$$

;

$$x \to 2$$

;

$$t \to 0$$

$$dx = dt$$

$$x \to \infty$$

;

$$t \to \infty$$

$$I = \int_{0}^{\infty} e^{-t^2/30} \cdot dt = \frac{\sqrt{30\pi}}{2}$$

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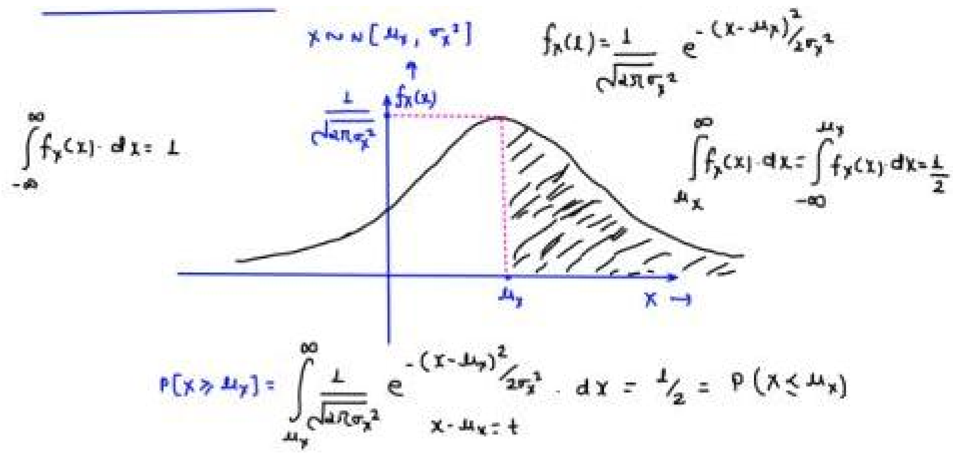
- Important Result:



Q

$$x \sim N(\mu_x, \sigma_x^2)$$

$$f_x(s) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{-(s-\mu_x)^2/2\sigma_x^2}$$

$$\text{Find } P(x > a) = ?$$

b

$$P(x > a) = \int_a^\infty \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{-(s-\mu_x)^2/2\sigma_x^2} \cdot dx$$

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} \cdot dt$$

$$\frac{x-\mu_x}{\sigma_x} = t \Rightarrow dx = \sigma_x dx$$

$$x = a \to t = \frac{a-\mu_x}{\sigma_x} \quad ; \quad x = \infty \to t = \infty$$

$$P(x > a) = \int_{\frac{a-\mu_x}{\sigma_x}}^\infty \frac{1}{\sqrt{2\pi}\sigma_x} e^{-t^2/2} \cdot (\sigma_x dx) \Rightarrow P(x > a) = \int_{\frac{a-\mu_x}{\sigma_x}}^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \cdot dx$$

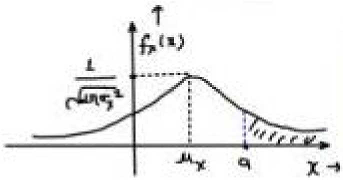
$$P(x > a) = \int_{\frac{a-\mu_x}{\sigma_x}}^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \cdot dt$$

$$P(x > a) = \Phi\left(\frac{a-\mu_x}{\sigma_x}\right)$$

$$x \sim N(\mu_x, \sigma_x^2)$$

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} \cdot dt$$

$$Q\left(\frac{a-\mu_x}{\sigma_x}\right) = \frac{1}{\sqrt{2\pi}} \int_{\frac{a-\mu_x}{\sigma_x}}^\infty e^{-t^2/2} \cdot dx$$



$$P(x > a) = \int_a^\infty f_x(s) \cdot dx = \Phi\left[\frac{a - \mu_x}{\sigma_x}\right]$$

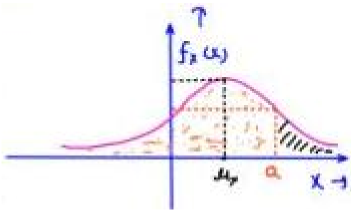
$$P(x > a) + P(x \leq a) = 1$$

- Conclusion:

For Gaussian Random variable (GRV) x: $N(\mu_x, \sigma_x^2)$

$$P(x > a) = \Phi\left(\frac{a - \mu_x}{\sigma_x}\right)$$

$$P(x \leq a) = 1 - P(x > a) = 1 - \Phi\left(\frac{a - \mu_x}{\sigma_x}\right)$$



- COF of GRV: $x \sim N(\mu_x, \sigma_x^2)$

$$Q(x) + Q(-x) = 1$$

$$F_x(s) = P(x \leq s) = 1 - P(x > s)$$

$$Q(x) = 1 - Q(-s)$$

$$F_x(s) = 1 - \Phi\left(\frac{x - \mu_x}{\sigma_x}\right) = \Phi\left(\frac{\mu_x - x}{\sigma_x}\right)$$

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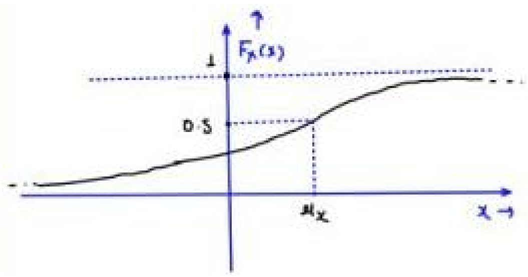


$$F_\lambda(x) = 1 - Q\left(\frac{x - \mu_x}{\sigma_x}\right)$$

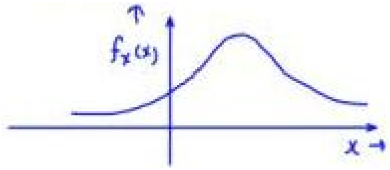
$$F_\lambda(\mu_x) = 1 - Q\left(\frac{\mu_x - \mu_y}{\sigma_x}\right)$$

$$= 1 - Q(0) = 1 - \frac{1}{2} = 0.5$$

$$F_\lambda(\mu_x) = P[x \leq \mu_x] = \frac{1}{2}$$

Q.

$$x \sim N(\mu_x, \sigma_x^2)$$



Find

$$\underbrace{P(a < x \leq b)}_{x:CRV}$$

;

$$P(x = \alpha) = 0$$

;

$$P(a \leq x < b) = P(a \leq x \leq b) = ?$$

$$P(x < x \leq b) = P(a < x < b) = P(a \leq x < b) = P(a \leq x \leq b)$$

$$= F_\lambda(b) - F_\lambda(a)$$

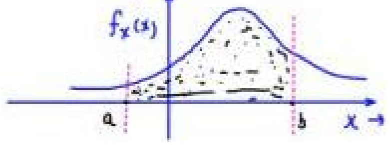
$$= 1 - Q\left(\frac{b - \mu_x}{\sigma_x}\right) - \left\{1 - Q\left(\frac{a - \mu_x}{\sigma_x}\right)\right\}$$

$$= P(x > a) - P(x > b)$$

$$= \boxed{Q\left(\frac{a - \mu_x}{\sigma_x}\right) - Q\left(\frac{b - \mu_x}{\sigma_x}\right)}$$

$$= \frac{Q\left(\frac{a-\mu_x}{\sigma_x}\right) - Q\left(\frac{b-\mu_x}{\sigma_x}\right)}{P(x - \mu_x) - Q\left(\frac{b-\mu_x}{\sigma_x}\right)}$$

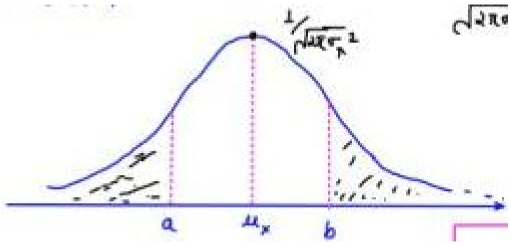
$$= Q\left(\frac{a - \mu_x}{\sigma_x}\right) - Q\left(\frac{b - \mu_x}{\sigma_x}\right)$$



Conclusion:-

$$N(\mu_x, \sigma_x^2)$$

$$f_\lambda(2) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{-(x-\mu_x)^2/2\sigma_x^2}$$



$$P(x > \mu_x) = P(x \leq \mu_x) = \frac{1}{2}$$

$$P(x \geq b) = Q\left(\frac{b - \mu_x}{\sigma_x}\right)$$

$$P(x \leq a) = 1 - P(x > a) = 1 - Q\left(\frac{a - \mu_x}{\sigma_x}\right)$$

$$P(a \leq x \leq b) = P(x > a) - P(x > b) = Q\left(\frac{a - \mu_x}{\sigma_x}\right) - Q\left(\frac{b - \mu_x}{\sigma_x}\right)$$

Other standard distribution:-

→ Exponential

→ Rayleigh

→ Gamma

→ Laplacian

→ Chi-square

→ Cauchy

→ Triangular

→ Bernoulli

→ Binomial

→ Poisson

4 Uniform

4 Gaussian | Normal

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Q.

$$x \sim \Delta(-1, 2, 4)$$

$$f_y(x) = \begin{cases} \int_{-\infty}^{\infty} f_x(x)\, dx = 1 \\ \frac{1}{2} x \le x < K = 1 \\ K = 2/5 \end{cases}$$

(i) Find

$$x$$

and mean of

$$x \le 2$$

(ii)

$$P(0 < x \le 2) =? = \frac{1}{2} x \left(\frac{2}{15} + \frac{2}{5} \right) \times 2 = \frac{8}{15}$$

(iii)

$$E[y] = \int_{-\infty}^{\infty} x f_y(x)\, dx = \int_{-1}^{4} x f_x(x)\, dx$$

$$= \int_{-1}^{2} x \left(\frac{2x+2}{15} \right) \cdot dx + \int_{2}^{4} x \left(-\frac{x}{5} + \frac{4}{5} \right) \cdot dx = -\frac{1+2+4}{3} = \frac{5}{3}$$

$$x \sim \Delta[a, b, c]$$

$$\checkmark \text{mean } E[x] = \frac{a+b+c}{3}$$

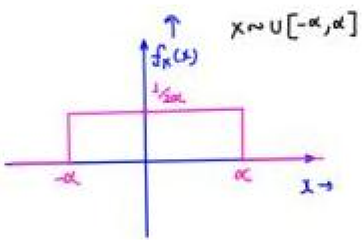
$$\text{or } \sigma_x^2 = E[x^2] - \{E[x]\}^2$$

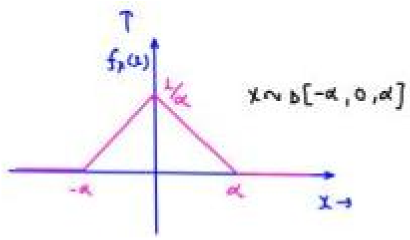
(b)

$$P(0 < x \le 2) = \int_{0}^{2} f_x(x)\, dx = \int_{0}^{2} \frac{2}{15}(x+1)\, dx$$

$$= \frac{2}{15} \left[\frac{x^2}{2} + x \right]_0^2 = \frac{2}{15}[2+2] = \boxed{\frac{8}{15}}$$

Important Results:





4 mean (E[x] = -\frac{\alpha + \alpha}{2} = 0)
4 MSV (E[x^2] = \alpha*3^2)*
4 Variance (\sigma{\lambda}^{2} = \frac{(2\alpha)^{2}}{12} = \frac{\alpha^{2}}{3})

4 Mean (E[x] = -\frac{\alpha + 0 + \alpha}{3} = 0)
4 MSV (E[x^2] = \alpha*6^2)*
4 Variance (\sigma{\lambda}^{2} = \frac{\alpha^{2}}{6})

Q.

$$f_x(x) = \begin{cases} \frac{x}{3}e^{-x^2/6} & ; x > 0 \\ 0 & ; 0/\infty \end{cases}$$

Find mean ? ; Answer:

$$\sqrt{\frac{3\pi}{2}}$$

$$E[x] = \int_{-\infty}^{\infty} x f_x(x)\, dx$$

$$= \int_0^{\infty} x \cdot \frac{x}{3} e^{-x^2/6} \cdot dx = \int_0^{\infty} \frac{x^2}{3} e^{-x^2/6} \cdot dx = ? = \frac{3}{2} x \sqrt{\frac{6\pi}{3}} = \sqrt{6\pi/2} = \sqrt{\frac{3\pi}{2}}$$

what we remember

$$f_y(y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{-(y-\mu_y)^2/2\sigma_y^2} \quad ; \mu_y = 0$$

$$\sigma_y^2 = 3$$

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$$f_Y(y) = \frac{1}{\sqrt{6\pi}} e^{-y^2/6} \quad ; \quad \begin{matrix} \mu_Y = 0 \\ \sigma_y^2 = 3 \end{matrix}$$

$$E[Y^2] = \int_{-\infty}^{\infty} y^2 f_Y(y)\, dy = \sigma_Y^2 + \mu_Y^2$$

$$\int_{-\infty}^{\infty} y^2 \times \frac{1}{\sqrt{6\pi}} e^{-y^2/6} \cdot dy = 3 + 0$$

$$\int_{-\infty}^{\infty} \underbrace{y^2 e^{-y^2/6}}_{\text{even}} \cdot dy = 3\sqrt{6\pi} \quad \int_{0}^{\infty} x^2 \cdot e^{-x^2/6} \cdot dx = \frac{3}{2}\sqrt{6\pi}$$

$$Q. \quad I = \int_{-\infty}^{\infty} (x+3)^2 \cdot e^{-3(x+3)^2} \cdot dx = ?$$

4

$$I + 3 = t$$

$$dx = at$$

$$\int_{-\infty}^{\infty} x^2 f_X(x)\, dx$$

$$I = \int_{-\infty}^{\infty} t^2 e^{-3t^2} \cdot dt$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X^2} e^{-(x-\mu_X)^2/2\sigma_X^2}$$

$$I = \frac{1}{6}\sqrt{\frac{\pi}{3}}$$

$$f_X(x) = \sqrt{\frac{3}{\pi}} e^{-3x^2}$$

$$\mu_X = 0$$

$$2\sigma_X^2 = 1/3$$

$$\sigma_X^2 = 1/6$$

$$\int_{-\infty}^{\infty} x^2 \times \sqrt{\frac{3}{\pi}} e^{-3x^2} \cdot dx = 1/6 + 0$$

Q

$$f_X(x) = \frac{1}{\sqrt{\pi}} e^{-(x-0.5)^2}$$

$$; f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X^2} e^{-(x-\mu_X)^2/2\sigma_X^2}$$

Given that

$$P(X > L) = 0.16$$

$$\mu_X = 0.5$$

$$\sigma_X^2 = 1/2$$

Find (a)

$$P(0.5 < X < L)$$

(b)

$$P(X < 0.5)$$

(c)

$$P(X \leq 0)$$

(d)

$$F_X(L)$$

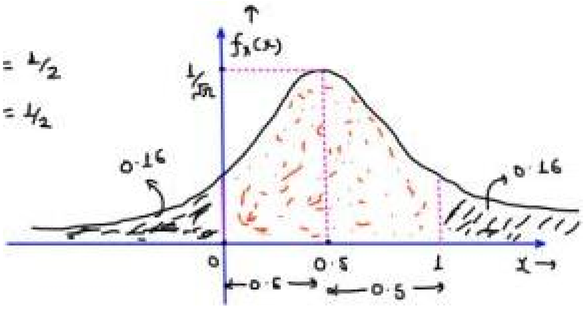
(e)

$$P(|X - 0.5| < 0.5)$$

$$P(X > 0.5) = 1/2$$

(ii)

$$P(X < 0.5) = 1/2$$



(i)

$$P(0.5 < X < L) = 1/2 - 0.16 = 0.34$$

$$|X - 0.5| < 0.5$$

(iii)

$$P(X \leq 0) = 0.16$$

$$-0.5 < X - 0.5 < 0.5$$

(iv)

$$F_X(L) - P(X \leq L) = 1 - 0.16 = 0.84$$

$$0 < X < L$$

(v)

$$P(0 < X < L) = 1 - (0.16 + 0.16) = 0.68$$

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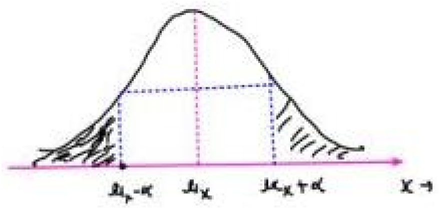


$$P(x > \mu_x + \alpha) = P = P(x < \mu_x - \alpha)$$

Transformation of Random Variable:

(i) Linear Transformation:

PDF of R.V. X

A R.V. X is given. 4 f_P(X) is given.

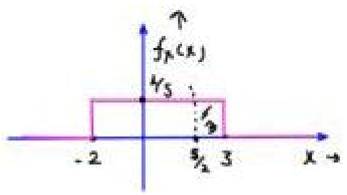
Another Random variable Y is defined as

Y = ax + b; a & b are some constant

4 Linear Transformation doesn't change the nature of R.V.

$$x \sim \text{Uniform} \quad \xrightarrow[Y=ax+b]{\text{Linear Transformation}} \quad Y \sim \text{Uniform}$$

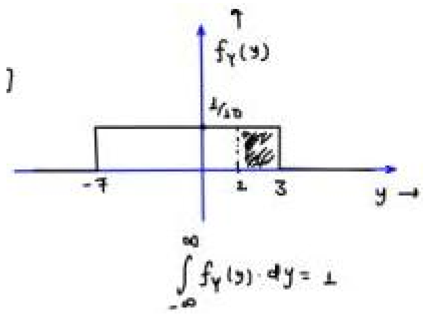
$$x \sim \text{Gaussian} \quad \xrightarrow[Y=ax+b]{\text{Linear Transformation}} \quad Y \sim \text{Gaussian}$$



Q. x ~ u [-2, 3]

$$Y = 2x - 3$$

(i) find the PDF of (Y).
(ii) find (E[Y] = ?)
(iii) find (P[Y > 2] = ?)



(ii) ($E[Y] = E[2x - 3] = 2E[x] - 3 = 2\left(\frac{3 - 2}{2}\right) - 3 = -2$)
(ii) ($P[Y > 2] = P[2x - 3 > 2] = P[X > 5/2] = ?$)

$$= (3 - 5/2) \times 1/6 = \boxed{\frac{1}{10}}$$

$$Y = 2x - 3 \quad ; \quad x \sim U[-2, 3]$$

$$Y_{min} = 2(-2) - 3 \quad \Downarrow$$
$$= -7 \qquad\qquad Y \sim \text{Uniform}$$

$$Y_{max} = 2(3) - 3 = 3$$

$$Y \sim U[-7, 3]$$

$$E[Y] = \frac{3 - 7}{2} = -2$$

$$P[Y > 2] = \frac{1}{10} \times 1 = 0.1$$

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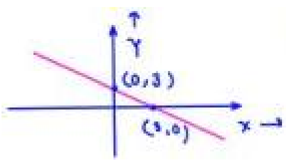
Q. $x \sim N(2, 4)$

$\mu_x = 2, \sigma_x^2 = 4; \sigma_x = 2$

another R.V. Y is defined. The relation b/w x and Y is given



by the following graph

$f_x(x) = \frac{1}{\sqrt{8\pi}} e^{-(x-2)/8}$

(i) Find PDF of Y.

(ii) Find $P(-1 < Y < 8) =?$

$\rightarrow Y = -x + 3$

$P(-1 < Y < 8) = P(-1 < -x + 3 < 3)$

$\rightarrow P(x > 0) = Q\left(\frac{a - \mu_x}{\sigma_x}\right)$

$= P(-4 < -x < 0)$

$= P(0 < x < 4) = P(x > 0) - P(x > 4) = Q\left(\frac{0-2}{2}\right) - Q\left(\frac{4-2}{2}\right) = Q(-1) - Q(1)$

$Y = -x + 3$

$\mu_y = -\mu_x + 3$

$\boxed{\mu_y = -2 + 3 = 1}$

$\sigma_y^2 = (-1)^2 \sigma_x^2$

$\sigma_y < 2; \quad \boxed{\sigma_y^2 = 4}$

$x \sim \text{Gaussian}$

$Y \sim \text{Gaussian}$

$x \sim N(\mu_x, \sigma_x^2)$

$\mu_x = 2$

$\sigma_x^2 = 4$

$\boxed{Y \sim N(\mu_y, \sigma_y^2); \quad Y \sim N(1, 4)}$

$f_y(y) = \frac{1}{\sqrt{8\pi}} e^{-(y-2)/8}$

$P(-1 < Y < 3) = P(Y > -1) - P(Y > 3)$

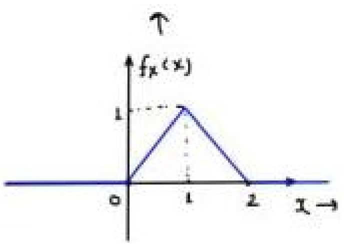
$= Q\left(\frac{-1-1}{2}\right) - Q\left(\frac{3-1}{2}\right)$

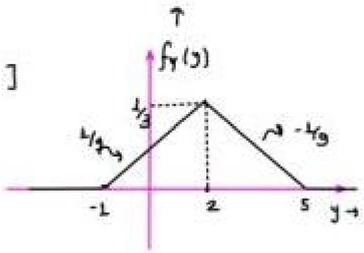
$\boxed{P(-1 < Y < 3) = Q(-1) - Q(1)}$

Q

$y : \text{CRV}$

$$f_x(x) = \begin{cases} x & ;\ 0 \le x < 1 \\ 2 - x & ;\ 1 \le x \le 2 \\ 0 & ;\ 0/\omega \end{cases}$$





$\int_{\text{CRV}}^{Y=3x-1}$

(a) Sketch $f_y(y)$ and define it.

(b) Find $E[Y]=?$

$\rightarrow (b) \quad E[Y]=E[3x-1]=3E[x]-1$

$=3\left[\frac{0+1+2}{3}\right]-1=\boxed{2=E[Y]}$

$x \sim \Delta[0,1,2]$

$E[x]=\frac{0+1+2}{3}=1$

$Y=3x-1 \quad x \in [0,2]$

$Y_{\min}=\delta(0)-1=-1$

$Y_{\max}=3(2)-1=5 \quad y \in [-1,5]$

when $x=1; \quad y=3(1)-1$

$=2$

$Y \sim \Delta[-1,2,5]$

$\frac{1}{2} \times 6 \times k=1$

$E[Y]=\frac{-1+2+5}{3}$

$\sqrt{E[Y]=2}$

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$$f_Y(y)=\begin{cases}\frac{y+1}{3} & ;-1 \leq y<2 \\ \frac{s-y}{3} & ;2 \leq y \leq 5 \\ 0 & ;0/\omega\end{cases}$$

Q. Let $x \in \{0,1\}$ represent a Binary R.V. indicating the bit sent in a digital communication system.

$$P_x(0)=0.7 \quad ; \quad P_x(1)=0.3 \quad E[x]=0(0.7)+1(0.3)=0.3$$

Let the modulator of signal be-

$$\text{DRV} \quad S=2x-1$$

$$N \sim N(0,1)$$

(a) Find the PMF of S.

(b) If Received signal is $R = S + N$, where N is zero mean, unit variance Gaussian noise. Find expected value of R.

$x \in \{0,1\} \quad ; \quad P(X=0) = 0.7 \quad R = S + N$

$\qquad\qquad\qquad\qquad P(X=1) = 0.3 \quad E[R] = E[S] + E[N]$

$$E[S] = E[2x - 1]$$

$$S = 2x - 1$$

$$= 2E[X] - 1$$

$$S = 2(0) - 1 = -1$$

$$= 2(0.3) - 1$$

$$S = 2(1) - 1 = 1$$

$$E[S] = -0.4$$

$$P(S = -1) = P(X = 0) = 0.7$$

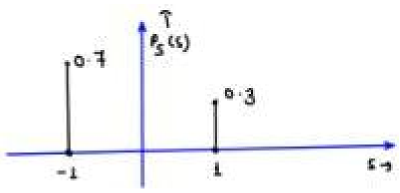
$$P(S = 1) = P(X = 1) = 0.3$$

$$x \in \{0, 1\}; E[x] = 0.3$$

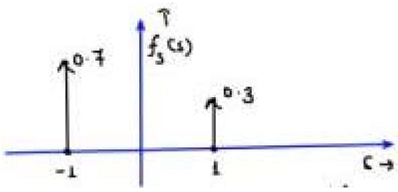
$$S \in \{-1, 1\}$$

$$E(S) = -1(0.7) + 1(0.3)$$

$$E(1) = -0.4$$



$$P_S(s) = 0.7\delta[S+1] + 0.3\delta[S-1]$$



$$f_S(s) = 0.7\delta(s+1) + 0.3\delta(s-1)$$

- DRV to DRV Transformation:-

Q. $C_{X \in \{-2,-1,0,1,2\}}$

$$P_X(x) = 1/S \quad ; \quad \text{for each } x$$

R.V. Y is defined as, $Y = x^2$

(a) Find the PMF of Y.

L(b) Find variance of Y.

$$Y = x^2$$

$$\sigma_Y^2 = ?$$

$$Y = ax + b$$

$$\sigma_Y^2 = a^2\sigma_X^2$$

$$Z = ax + bY$$

$$\sigma_Z^2 = a^2\sigma_Y^2 + b^2\sigma_Y^2 + 2ab\sigma_Y$$

$$P(X=-2) = P(X=-1) = P(X=0) = P(X=1) = P(X=2) = 1/S$$

$$Y = x^2$$

$$x = -2; \quad Y = 4 \quad x = -1; \quad Y = 1 \quad ; \quad x = 0; \quad Y = 0$$

$$x = 2; \quad Y = 4 \quad x = 1; \quad Y = 1$$

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$$\begin{aligned}
&\text{ORV} \quad \gamma \in \{0,1,4\} \\
&P[Y=0] = P[X=0] = 1/5 \\
&P[Y=1] = P[X=-1] + P[X=1] = 2/5 \\
&P[Y=4] = P[X=-2] + P[X=2] = 2/5 \\
&f_Y(y) = \tfrac{1}{5}\delta(y) + \tfrac{2}{5}\delta(y-1) + \tfrac{2}{5}\delta(y-4) \\
&\sigma_Y^2 = E[Y^2] - \{E[Y]\}^2 \quad ; \quad E[Y^2] = (0)^2 1/5 + (1)^2(2/5) + (4)^2(3/5) = 2/5 + \tfrac{32}{5} \\
&= \tfrac{94}{5} - 4 \\
&\sigma_Y^2 = \tfrac{14}{5} \quad E[Y] = 0(1/5) + 1(2/5) + 4(3/5) = \tfrac{10}{5} = 2
\end{aligned}$$

Q. Let $x, z \in \{0,1\}$ be independent ORVs with:-

$$P_x(0) = 0.6 \quad ; \quad P_x(1) = 0.4$$

$$P_z(0) = 0.7 \quad ; \quad P_z(1) = 0.3$$

$$\text{or } x \in \mathbb{R}$$

$$Y = x \oplus z$$

(a) Find PMF of Y.

(b) Find $E[Y]$

$$y \in \{0,1\}$$

$$z \in \{0,1\}$$

$$Y = x \oplus z$$

$$= 0 \oplus 0 = 0\checkmark$$

$$= 0 \oplus 1 = 1$$

$$= 1 \oplus 0 = 1$$

$$= 1 \oplus 1 = 0$$

$$Y = x \oplus z$$

$$E[Y] = \boxed{E[y \oplus z]} \quad x\times$$

$$= E[y] \oplus E[z] \quad x$$

$$Y \in \{0,1\}$$

$$P[Y=0] = P[X=0, z=0] + P[X=1, z=1]$$

$$= P[X=0] \cdot P[Z=0] + P[Y=1] \cdot P[Z=1]$$

$$= 0.6 \times 0.7 + 0.4 \times 0.3$$

$$P[Y=0] = 0.42 + 0.12 = 0.54$$

$P[Y=0] = 0.54 \quad Y \in \{0,1\}$
$P[Y=1] = 0.46 \quad P[Y=1] = P[X=0] \cdot P[Z=1] + P[X=1] \cdot P[Z=0]$
$= 0.6 \times 0.3 + 0.7 \times 0.4$
$= 0.18 + 0.28 = 0.46$
$E[Y] = (0)(0.54) + (1)(0.46)$
$E[Y] = 0.46 \quad \text{and}$

Q. Let $x \in \{1,2,3,4\}$ with $P(x=1) = 3/8$; $P(x=2) = 1/4$

$$Y = \begin{cases} x & ;\ \text{even} \\ 5-x & ;\ \text{odd} \end{cases}$$

$$P(X=3) = 1/8$$

$$P(X=4) = 1/4$$

Find $F_Y(4) - F_Y(2)$

$$P[Y \le 4] - P[Y \le 2] = P[2 < Y \le 4]$$

$$x \in \{1,2,3,4\}$$

$$\downarrow \quad \downarrow \text{odd} \quad \text{odd}$$

$$x = 1 \to \text{odd} \quad Y = 5 - 1 = 4$$

$$\to y = 2 \to \text{Even} \quad Y = 2$$

$$Y \in \{4,2,2,4\}$$

$$\to x = 3 \to \text{odd} \quad Y = 5 - 3 = 2$$

$$x = 4 \to \text{Even} \quad Y = 4$$

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$$y \in \{1,4\}$$

$$P[Y=2] = P[x=2] + P[x=3] = 1/8 + \frac{1}{4} = 3/8$$

$$P[Y=4] = 5/8$$

$$P[Y=4] = P[x=4] + P[x=1]$$

$$= 3/8 + 1/4 = 5/8$$

$$F_Y(4) = P[Y \leq 4]$$

$$= P[Y=2] + P[Y=4] = 1$$

$$F_Y(2) = P[Y \leq 2]$$

$$= P[Y=2] = 3/8$$

$$F_Y(4) - F_Y(2) = 1 - 3/8 = \boxed{5/8}\ \text{plus}$$

Q. Let $x_1, x_2 \in \{1,2,3\}$ be identical and independent (i.i.d. R.V.)

Random variable with

$$P(X) = \begin{cases} 0.2 & ; x=1 \\ 0.5 & ; x=2 \\ 0.3 & ; x=3 \end{cases}$$

$$P(x_1 = 1) = P(x_2 = 1) = 0.2$$

$$P(x_1 = 2) = P(x_2 = 2) = 0.5$$

$$P(x_1 = 3) = P(x_2 = 3) = 0.3$$

$$\text{RV } M = \max(x_1, x_2)$$

$$\text{RV } M = \min(x_1, x_2)$$

$$(5) P\left(\frac{M=3}{M=3}\right) = \frac{P[M=3 \cap M=3]}{P(M=3)}$$

(a) PMF of M and M

$$= P\left[\frac{x_1 = 3, x_2 = 3}{0.09}\right]$$

(b) Are M and M independent? $\rightarrow$ NO

$$= \frac{0.3 \times 0.3}{0.09} = 1$$

(c) $P(M = 3/M = 3) = ? = 1 \neq P(M=3)$

4. $M \in \{1,2,3\}$ x_1 and x_2 independent

$$P(M=1) = P(x_1 = 1, x_2 = 1) = P(x_1 = 1) \cdot P(x_2 = 1) = (0.2)^2 = 0.04$$

$$P(M=2) = P(x_1 = 1, x_2 = 2) + P(x_1 = 2, x_2 = 2) + P(x_1 = 2, x_2 = 1)$$

$$= (0.2 \times 0.5) + (0.5 \times 0.5) + (0.5 \times 0.2)$$

$$= 0.1 + 0.25 + 0.1 = 0.45$$

$$P(M = 3) = 1 - 0.04 - 0.45 = 0.51$$

4. $M \in \{1, 2, 3\}$

$$P(M = 3) = P(x_1 = 3, x_2 = 3) = 0.3 \times 0.3 = 0.09$$

$$P(M = 2) = P(x_1 = 3, x_2 = 1) + P(x_1 = 2, x_2 = 3) + P(x_1 = 2, x_2 = 2)$$

$$= 0.3 \times 0.5 + 0.5 \times 0.3 + 0.5 \times 0.5 = 0.55$$

$$P(M = 1) = 1 - 0.09 - 0.55 = 0.36$$

$$M \in \{1, 2, 3\}$$

$$P[M = \infty] \in \{0.04, 0.45, 0.51\}$$

$$M \in \{1, 2, 3\}$$

$$P[M = \infty] = \{0.36, 0.55, 0.09\}$$

4	x_1	x_2	$M = \max(x_1, x_2)$	$M = \min(x_1, x_2)$
	1	1	1 $\vee$	1
	1	2	2 $\vee$	1
	1	3	3	1
	2	1	2 $\vee$	1
	2	2	2 $\vee$	2
	2	3	3	2
	3	1	3	1
	3	2	3	2
	3	3	3 $\rightarrow \vee$	3 $\rightarrow \vee$

$$x_1, x_2 \in \{1, 2, 3\}$$

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4 CRV to DRV Transformation:-

$$\nu_{DRV} \rightarrow DRV$$

$$\nu_{DRV} \rightarrow CRV$$

$$\nu_{CRV} \rightarrow DRV$$

Q:

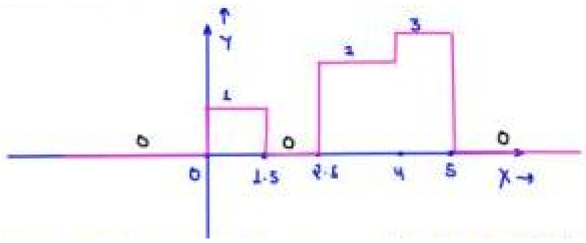
$$x \sim U[0, 5]$$

Another R.V. Y is defined.

Relation b/w R.V. X and Y is given by:-

$$\underbrace{CRV \rightarrow CRV}_{\Downarrow}$$

$$Y = ax + b\vee$$



(i) Find the expression of

$$f_Y(y)$$

(ii) Find

$$E[Y] = ? = 1.5$$

4

$$\underset{D.R.V.}{\rightarrow} Y \in \{0, 1, 2, 3\}$$

$$E[Y] = 0 + 0.3 + 0.6 + 0.6 = \boxed{1.5}$$

$$P[Y = 0] = P[1.5 < x < 2.5] + P[x \neq 0.5] + P[x \neq 0]$$

$$= 1 \times 1/3 +$$

$$P[Y = 0] = 0.2$$

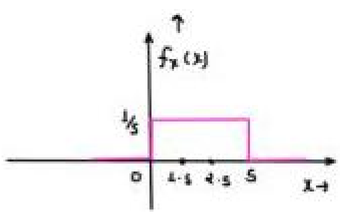
$$P[Y = 1] = P[0 < x < 1.5] = 1/5 \times 1.5 = 0.3$$

$$P[Y = 2] = P[2.5 < x < 4] = 1/5 \times 1.5 = 0.3$$

$$P[Y = 3] = P[4 < x < 5] = 1/5 \times 1 = 0.2$$

$$Y \in \{0, 1, 2, 3\} \quad P[Y = y_i] = \{0.2, 0.3, 0.3, 0.2\}_{i=0}^{3}$$

$$f_Y(y) = 0.2\delta(y) + 0.3\delta(y-1) + 0.3\delta(y-2) + 0.2\delta(y-3)$$



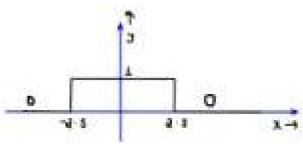
A uniformly distributed random variable X with probability density function

$$f_X(x) = \frac{1}{10}(u(x+5) - u(x-5))$$

where

$$u(\cdot)$$

is the unit step function, is passed through a transformation given in the figure below.



The probability density function of the transformed random variable Y would be:

Options:

(A)

$$f_Y(y) = \frac{1}{5}(u(y+2.5) - u(y-2.5))$$

(B)

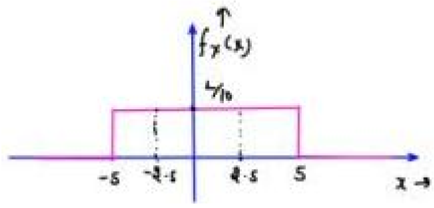
$$f_Y(y) = 0.5\delta(y) + 0.5\delta(y-1)$$

(C)

$$f_Y(y) = 0.25\delta(y+2.5) + 0.25\delta(y-2.5) + 0.5\delta(y)$$

(D)

$$f_Y(y) = 0.25\delta(y+2.5) + 0.25\delta(y-2.5) + \frac{1}{10}(u(y+2.5) - u(y-2.5))$$



$$Y \in \{0, 1\}$$

$$P[Y=0] = P[x > 2.5] + P[x < -2.5] = \frac{2.5}{10} + \frac{2.5}{10} = 1/2 = 0.5$$

$$P[Y=1] = P[-2.5 < x < 2.5] = 5 \times 1/10 = 0.5$$

$$f_Y(y) = 0.5\delta(y) + 0.5\delta(y-1)$$

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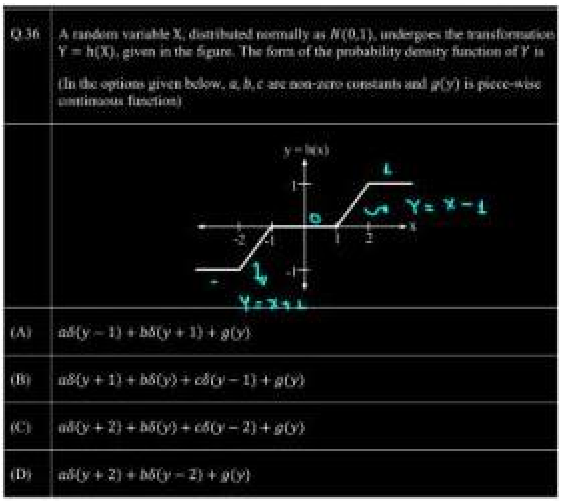

Q.36	A random variable X, distributed normally as $N(0,1)$, undergoes the transformation $Y = h(X)$, given in the figure. The form of the probability density function of Y is (In the options given below, a, b, c are non-zero constants and $g(y)$ is piece-wise continuous function)
(A)	$a\delta(y-1) + b\delta(y+1) + g(y)$
(B)	$a\delta(y+1) + b\delta(y) + c\delta(y-1) + g(y)$
(C)	$a\delta(y+2) + b\delta(y) + c\delta(y-2) + g(y)$
(D)	$a\delta(y+2) + b\delta(y-2) + g(y)$

$$f_Y(y) = ?$$

$$x \sim N(0, L)$$

$$\mu_Y = 0$$

$$\sigma_Y^2 = 1$$

$$\downarrow Y \in \{-1, 0, L\} + (0, L) + (-1, 0)$$

$$\text{Mixed RV}$$

$$Y = \begin{cases} 0 & ; -1 < x < 1 \\ 1 & ; x > 2 \\ -1 & ; x < -2 \\ x-1 & ; 1 < x < 2 \\ x+1 & ; -2 < x < -1 \end{cases}$$

$$P[Y=0] = P[-1 < x < L] = P[Y > -L] - P[X > L]$$

$$= Q\left(\frac{-1}{L}\right) - Q\left(\frac{1}{L}\right) = Q(-1) - Q(1) = \text{constant} = b$$

$$P[Y=L] = P[X > 2] = Q\left(\frac{2-0}{L}\right) = Q(2) = \text{constant} = c$$

$$P(Y=-L) = P(X < -2) = 1 - P(X > -2)$$

$$= 1 - Q\left(-\frac{2}{L}\right) = Q(1) = \text{constant} = a$$

$$1 < x < 2 \quad \text{Gaussian}$$

$$\downarrow Y = x - L$$

$$\text{Gaussian}$$

$$E[Y] = E[Y] - L$$

$$= 0 - 1 = -L$$

$$\sigma_Y^2 = \sigma_X^2 = 1$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-(y+1)/2}; \quad 0 < y < L$$

$$-2 < y < -1 \quad \text{Gaussian}$$

$$\downarrow Y = x + L$$

$$\text{Gaussian}$$

$$E[Y] = 1; \quad f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-(y-1)/2}; \quad -1 < y < 0$$

$$\sigma_Y^2 = 1$$

$$\text{POF} \quad f_Y(y) = \begin{cases} b\delta(y) & ; y = 0 \\ a\delta(y+1) & ; y = -1 \\ c\delta(y-1) & ; y = 1 \\ p(y) & ; -1 < y < 0 \\ Q(y) & ; 0 < y < 1 \end{cases}$$

$$f_Y(y) = a\delta(y+1) + b\delta(y) + c\delta(y-1) + g(y)$$

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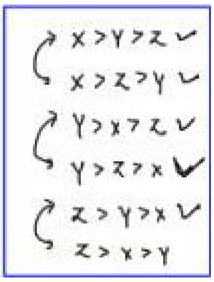
N.B. We will see more examples of CRV to CRV Non-linear Transformation, but those are not much important. We can see those examples after the next lecture.

For x, y, z are 3 i.i.o. R.V.

Find the probability such that R.V. x is largest.

4



4

$$P(Z \text{ is smallest}) = 1/3$$
$$P(X \text{ is smallest}) = 2/6 = 1/3$$
$$P(Y \text{ is smallest}) = 1/3$$

$$x \sim U[0, L]$$

$$Y \sim U[0, L]$$

$$Z \sim U[0, L]$$

$$P(X \text{ is largest}) = \frac{2}{6} = \frac{1}{3}$$

$$P(Y \text{ is largest}) = \frac{2}{6} = 1/3$$

$$P(Z \text{ is largest}) = 2/6 = 1/3$$

4

$$x_1, x_2, x_3 \ldots x_n$$

are n i.i.o. R.V.

$$P(x_1 \text{ is largest}) = 1/n$$

$$P(x_2 \text{ is largest}) = 1/n$$

$$P(x_n \text{ is largest}) = 1/n$$

$$P(x_1 \text{ is smallest}) = 1/n$$

$$P(x_2 \text{ is smallest}) = 1/n$$

$$P(x_n \text{ is smallest}) = 1/n$$

$$P(X_1 > X_2 > X_3 \ldots X_n) = \frac{1}{n!}$$

$$\overline{x}_n \quad \overline{y}_{n-1} \quad \overline{y}_{n-2} \quad \overline{y}_{n-3}$$

Q. 4 R.V. x, y, z, w are i.i.d. R.V.

$$(a) P(X \text{ is largest}) = 1/4 = \frac{6}{24}$$

$$(b) P(X > Y > Z > W) = 1/4! = \frac{1}{24}$$

$$\begin{array}{ccccc} \hline 3 & 3 & 3 & 3 & = 4! \\ 4 & 3 & 2 & 1 & \end{array}$$

$$4x > y > z > w \quad x > z > w > Y$$

$$x > y > w > z \quad x > w > z > Y$$

$$x > z > y > w \quad x > w > y > z$$

- Some miscellaneous Questions:-

Q. Given that

$$\max(x, y) > z$$

; which of the following statement is true?

$$(a) x > z, y < z \quad (b) x < z, y > z$$

$$(c) x > z, y > z \quad (d) all$$

Q. Given that

$$\max(x, y) < z$$

; which of the following statement is true?

$$(a) x < z, y < z \quad (b) x > z, y < z$$

$$(c)x < z, y > z \quad (d)x > z, y > z$$

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Q. x and Y are two independent CRV.

$$P(x > t_{1/2}) = 0.3, \quad P(Y > t_{1/2}) = 0.4$$

$$\text{Find } P[\max(x, Y) > t_{1/2}]$$

$$P(x < t_{1/2}) = 0.7$$

$$P(Y < t_{1/2}) = 0.6$$

$$P[\max(x, Y) > t_{1/2}] = P[x > t_{1/2}, Y > t_{1/2}] + P[x < t_{1/2}, Y > t_{1/2}]$$

$$+P[x > t_{1/2}, Y < t_{1/2}]$$

$$= 0.8 \times 0.4 + 0.7 \times 0.4 + 0.3 \times 0.6$$

$$= 0.12 + 0.28 + 0.18 = \boxed{0.58}$$

$$P[\max(x, Y) > t_{1/2}] = L - P[\max(x, Y) < t_{1/2}]$$

$$= L - P[x < t_{1/2}, Y < t_{1/2}]$$

$$= 1 - 0.7 \times 0.6 = L - 0.42 = 0.58$$

Q. Given that $\min(x, y) > z$; which of the following statement is true?

$$(a)x > z, y < z$$

$$(b)x < z, y > z$$

$$(c)x > z, y > z$$

$$(d)x < z, y < z$$

Q. Given that $\min(x, y) < z$; which of the following statement is true?

$$(a)x < z, y < z$$

$$(b)x > z, y < z$$

$$(c)x < z, y > z$$

$$(d) \text{ all}$$

Q. x and Y are two independent CRV.

$$P(x > t_{1/2}) = 0.3, \quad P(Y > t_{1/2}) = 0.4$$

$$\text{Find } P[\min(x, Y) < t_{1/2}]$$

$$P[\min(x, Y) < t_{1/2}] = P[x < t_{1/2}, Y < t_{1/2}] + P[x < t_{1/2}, Y > t_{1/2}]$$

$$+P[x > t_{1/2}, Y < t_{1/2}]$$

$$= 0.7 \times 0.6 + 0.7 \times 0.4 + 0.3 \times 0.6$$

$$= 0.42 + 0.28 + 0.18 = 0.88$$

$$P[\min(x, Y) < t_{1/2}] = L - P[\min(x, Y) > t_{1/2}] = L - P[x > t_{1/2}, Y > t_{1/2}]$$

$$= L - 0.3 \times 0.4$$

$$= L - 0.12 = 0.88$$

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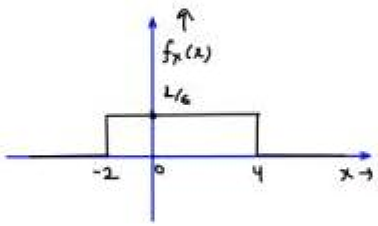

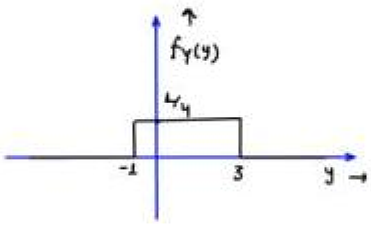

Q. x ~ u[-2, 4] ; y ~ u[-1, 3]

x & y are independent R.V.

e.v. M = max(x, y)

e.v. m = min(x, y)

Find the probabilities.

(a) p(m < 0)

(b) p(M > 0)

(c) p(-1 < m < 2)

(d) p(-1 < M < 2)

(e) p(-1 < m < 0 < M < 2)

Answers.

(a) 1/2 (b) 1 1/12 (c) 3/4

(d) 1/2 (e) 1/6

(a) p(m < 0) = p[min(x, y) < 0] = 1 - p[min(x, y) > 0]

= 1 - p[x > 0, y > 0] = 1 - p[x > 0] · p[y > 0]

= 1 - 2/3 x 3/4 = 1/2

(b) p(M > 0) = p[max(x, y) > 0] = 1 - p[max(x, y) < 0]

= 1 - p[x < 0, y < 0] = 1 - p[x < 0] · p[y < 0]

= 1 - 1/3 x 1/4 = 1 1/12

(s) p(-1 < m < 2) = p(m < 2) - p(m < -1) →

= p(m > -1) - p(m > 2)

= p[min(x, y) > -1] - p[min(x, y) > 2]

= p[x > -1, y > -1] - p[y > 2, y > 2]

= 5/6 x 1 - 1/3 x 1/4 = 5/6 - 1/12 = 3/12 = 3/4

(d) p(-1 < M < 2) = p(M < 2) - p(M < -1)

= p[max(x, y) < 2] - p[max(x, y) < -1]

= p[x < 2, y < 2] - p[x < -1, y < -1]

= 2/3 x 3/4 - 1/6 x 0 = 1/2

(e) p(-1 < m < 0 < M < 2) = p(-1 < m < 0, 0 < M < 2)

= p(-1 < m < 0) · p(0 < M < 2) X X

M and M depends on each other

= p[-1 < min(x, y) < 0, 0 < max(x, y) < 2]

= p[-1 < x < 0, 0 < y < 2] + p[0 < x < 2, -1 < y < 0]

= 1/6 x 1/2 + 1/3 x 1/4 = 1/12 + 1/12 = 1/6 ANS

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- Some Important Points:

(i) Addition of Independent R.V. $\rightarrow$ Uniform Gaussian
net $x_1 \sim u[0,1]$ Anything else

$x_2 \sim u[1,2]$

$x_3 \sim u[-2,3]$

$x = x_1 + x_2 + x_3 + \ldots\ldots\ldots x_n$

$f_x(x) = f_{x_1}(x) * f_{x_2}(x) * f_{x_3}(x) * \ldots \cdots * f_{x_n}(x)$

Q.

$x \sim u[-2, 2]$

$y \sim u[-3, 3]$

$p[x + 2y - 3 < 2]$

$p[x + 2y < 5]$

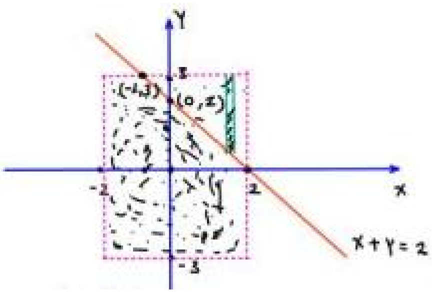
x and y are independent R.V. $p[-2 < x + y < 2]$

Find $p[x + y \leq 2]$, $p[x + 2y - 3 < 2]$, $p[|x + y| \leq 2]$

4

$f_x(x) = \begin{cases} \frac{1}{4}; & -2 < x < 2 \\ 0; & 0/\infty \end{cases}$ $f_y(y) = \begin{cases} \frac{1}{6}; & -3 < y < 3 \\ 0; & 0/\infty \end{cases}$

$f_{xy}(x, y) = f_y(x) f_y(y) = \begin{cases} \frac{1}{24}; & -2 < x < 2, -3 < y < 3 \\ 0; & 0/\infty \end{cases}$



$$\begin{aligned}
\text{(i)} \quad p(x + y \leq x) &= \int_{x=-1}^{2} \int_{y=-3}^{2-x} f_{xy}(x, y)\, dx \cdot dy + \int_{x=-2}^{-1} \int_{y=-3}^{3} f_{xy}(x, y)\, dx\, dy \\
&= 1 - \int_{x=-1}^{2} \int_{y=-3}^{3} f_{xy}(x, y)\, dx\, dy
\end{aligned}$$

$$\begin{aligned}
&= 1 - \int_{x=-1}^{2} \int_{y=1-x}^{3} \frac{1}{24}\, dx\, dy \\
&= 1 - \frac{1}{24} \int_{x=-1}^{2} 1 + x \cdot dx = 1 - \frac{1}{24} \left[x + \frac{x^2}{2} \right]_{-1}^{2} \\
&= 1 - \frac{1}{24} [2 + 2 - (-1 + 1/2)] \\
&= 1 - \frac{4 \cdot 5}{24} = \boxed{0.8125}
\end{aligned}$$

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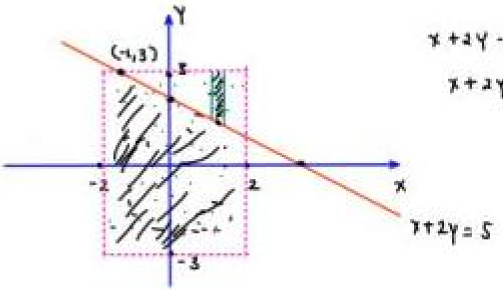
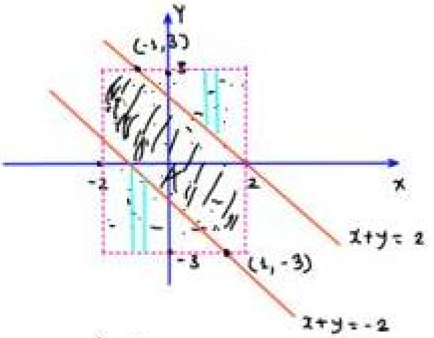


$x+2y-3<2$

$x+2y<5$



$$P[|x+y|<2] = 1 - \int_{x=-1}^{2}\int_{y=2-x}^{3} \frac{1}{24}\, dx\, dy - \int_{x=-2}^{x=1}\int_{y=-3}^{-2-x} \frac{1}{24}\, dx\, dy$$

$$= \boxed{0.625} \text{ Ans}$$

$$P(x+2y-3<2) = P(x+2y<5) = 1 - \int_{x=-1}^{2}\int_{y=-\frac{5-x}{2}}^{3} \frac{1}{24}\cdot dx\cdot dy$$

$$= 0.90625$$

$$x \sim U[-2,2]$$

$$z = x+y$$

$$z_{min} = -1\cdot 3 = -5$$

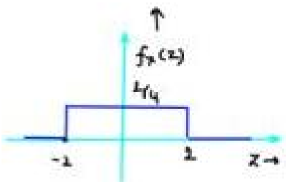
$$y \sim U[-3,3]$$

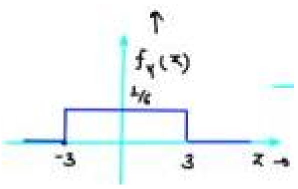
$$z_{max} = 2+3 = 5$$

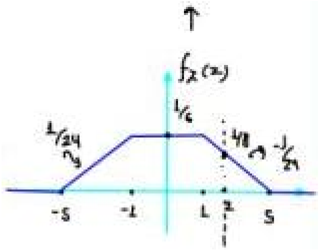
$$(9) \quad P[x+y\le 2] = P[z\le 2] = \int_{-\infty}^{2} f_x(z)\cdot dz$$

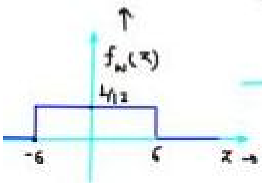
$$z = x+y$$

$$f_x(z) = f_y(z) * f_y(z)$$









$$P(z \le 2) = 1 - \frac{1}{2} \times 3 \times \frac{1}{8} = 1 - \frac{3}{16} = 0.8125$$

$$(3) \quad P[x + 2y - 3 < 2] = P[x + 2y < 5] = P(z < 5) = \int_{-\infty}^{5} f_z(z) \cdot dz$$

$$z = x + 2y$$

$$z = x + w$$

$$f_x(z) = f_y(z) * f_w(z)$$

$$\sim x \sim U[-1, 2]$$

$$y \sim U[-3, 3]$$

$$P(z < 5) = 1 - \frac{1}{2} \times 3 \times \frac{3}{48} = 1 - \frac{9}{96} = 0.90625$$

$$\sim w \sim U[-6, 6]$$

$$w = 2y$$

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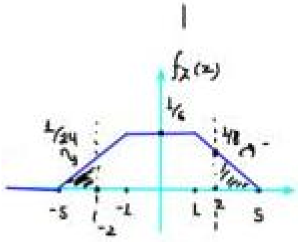
(5)

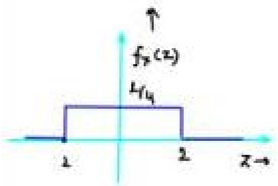
$$p[|x + y| < 2] = p[-2 < x + y < 2] = p(-2 < z < 2) = \int_{-1}^{2} f_z(z)\, dz$$

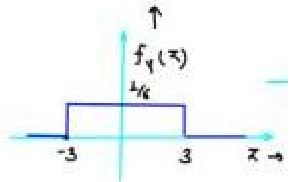
$$z = x + y$$

$$f_z(z) = f_x(z) * f_y(z)$$

$$p[|z| < 2] = 1 - \frac{2 \times 3}{16} = \boxed{0.625} \text{ Ans.}$$







4(ii) addition / subtraction of GRV will always be Gaussian only.

Q.

$$x \sim N(2, 4)$$

;

$$y \sim N(3, 2)$$

x and y are independent.

Find

$$p(x > \frac{3y}{2}) = ?$$

$$p(x > \frac{3y}{2}) = p(2x - 3y > 0) = p(z > 0) = Q\left(0 - \frac{\mu_z}{\sigma_z}\right)$$

$$f_z = 2x - 3y$$

$$\mu_z = 2\mu_x - 3\mu_y$$

$$= 2(1) - 3(3) = 2 - 9 = -7$$

$$\mu_x = -7$$

$$z = 2x - 3y$$

$$\sigma_z^2 = 4\sigma_x^2 + 9\sigma_y^2 - 12\sigma_z^2$$

$$\sigma_z^2 = 4\sigma_x^2 + 9\sigma_y^2$$

$$= 4(4) + 9(2)$$

$$\sigma_z^2 = 25$$

$$\sigma_z = 5$$

$$p(x > \frac{3y}{2}) = Q\left(\frac{7}{5}\right)$$

Ans.

$$z = ax + by$$

$$\sigma_z^2 = a^2\sigma_x^2 + b^2\sigma_y^2$$

$$+2ab\sigma_{xy}$$

$$cov(x, y)$$

x and y are independent

$$E[x\,y] = E[x] - E[y]$$

$$\sigma_{xy} = E[x\,y] - E[x] - E[y]$$

$$\sigma_{xy} = 0$$

Q.

$$x \sim N(0, 4)$$

;

$$y \sim N(0, 9)$$

$$\sigma_{xy} = 3$$

$$p(|x - y| > 2)$$

4 x and y are not independent

$$p(|z| > 2) = p(z > 2) + p(z < -2) = p(z > 2) + 1 - p(z > -2)$$

$$z = x - y$$

$$\mu_z + \mu_y - \mu_y = 0 - 0 = 0$$

$$\sigma_z^2 = \sigma_x^2 + \sigma_y^2 - 2\sigma_{xy} = 4 + 9 - 2(3) = 7$$

$$\boxed{\sigma_z = \sqrt{7}, \mu_z = 0}$$

$$= 1 + Q\left(\frac{2 - \mu_z}{\sigma_z}\right) - Q\left(\frac{-2 - \mu_z}{\sigma_z}\right)$$

$$= 1 + Q\left(\frac{2}{\sqrt{7}}\right) - Q\left(\frac{-2}{\sqrt{7}}\right)$$

$$= 2Q\left(\frac{2}{\sqrt{7}}\right)$$

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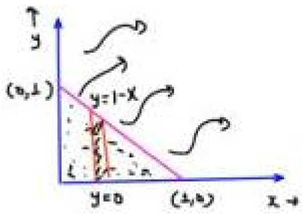
Q. $x \sim N(1,4)$; $y \sim N(2,9)$; $z \sim N(3,4)$

x, y, z are independent

$P\left(x - \frac{2y}{3}, z+2\right) = ?$

$P(3x - 2y > 3z + 6) = P(3x - 2y + 3z > 6)$

$= P(w > 6) = Q\left(\frac{6-\mu_w}{\sigma_w}\right) = Q\left(\frac{6-8}{\sqrt{108}}\right)$



$w = 3x - 2y + 3z$

$\mu_w = 3(1) - 2(2) + 3(5)$

$= 3 - 4 + 9 = 8$

$\sigma_w^2 = 9\sigma_x^2 + 4\sigma_y^2 + 9\sigma_z^2 = 36 + 36 + 36 = 108$; $\sigma_w = \sqrt{108}$

Q. x and y are two independent R.V.

$f_x(x) = 2e^{-2x}u(x)$; $f_y(y) = 3e^{-3y}u(y)$

another R.V. z is defined as $z = x + y$

(a) $P(z > 1) \rightarrow [\text{deg} : 0.5]$ (b) PDF of $z = ?$

$P(z > 1) = P(x + y > 1)$

$f_{xy}(x, y) = f_y(z)\ f_y(y) = 6e^{-2x}e^{-3y}\ \underbrace{u(x)}_{0<x<\infty}\underbrace{u(y)}_{0<y<\infty}$

$P(x + y > 1) = 1 - \int_{x=0}^{1}\int_{y=0}^{1-x} 6e^{-2x}e^{-3y}dxdy$

$= 1 - \frac{6}{3}\int_{x=0}^{1} e^{-2x}\left[e^{-3y}\right]_{1-x}^{0}$

$= 1 - \frac{6}{3}\int_{x=0}^{1} e^{-2x}\left[e^{-3y}\right]_{1-x}^{0}$

$= 1 - 2\int_{x=0}^{1} e^{-2x}\left[1 - e^{3x}e^{-3}\right]dx$

$= 1 - 2\int_{0}^{1} e^{-2x}dx + 2\int_{0}^{1} e^{-3}e^{x}dx$

$= 1 - \frac{2}{2}\left[e^{-2x}\right]_{1}^{0} + 2e^{-3}[e^{x}]_{0}^{1}$

$= 1 - \left[1 - e^{-2}\right] + 2e^{-3}\left[e^{1} \cdot 1\right] = +e^{-2} + 2e^{-2} - 2e^{-3}$

$= 3e^{-2} - 2e^{-3} = \boxed{0.8}$

$\underline{\underline{M - F}}$

$z = x + y$

$f_x(z) = f_x(x) * f_y(x)$

$f_x(x) = 2e^{-2x}u(x)\ f_y(x) = 3e^{-3x}u(x)$

$f_x(x) = 6e^{-2x}u(x) * e^{-3x}u(x)$

$= 6\left[\frac{1}{s+2} \cdot \frac{1}{s+3}\right]$

$= c\left[\frac{1}{s+2} - \frac{1}{s+3}\right] = 6\left[e^{-2x} - e^{-3x}\right]u(x)$

$f_x(x) = 6\left[e^{-2x} - e^{-3x}\right]u(x)$

$P(z > 1) = 6\int_1^\infty \left[e^{-2x} - e^{-3x}\right]dx = 6\left[\frac{e^{-2x}}{2} - \frac{e^{-3x}}{3}\right]_\infty^1$

$= 6\left[\frac{e^{-2}}{2} - \frac{e^{-3}}{3}\right] = 3e^{-2} - 2e^{-3} = \boxed{0.3}$

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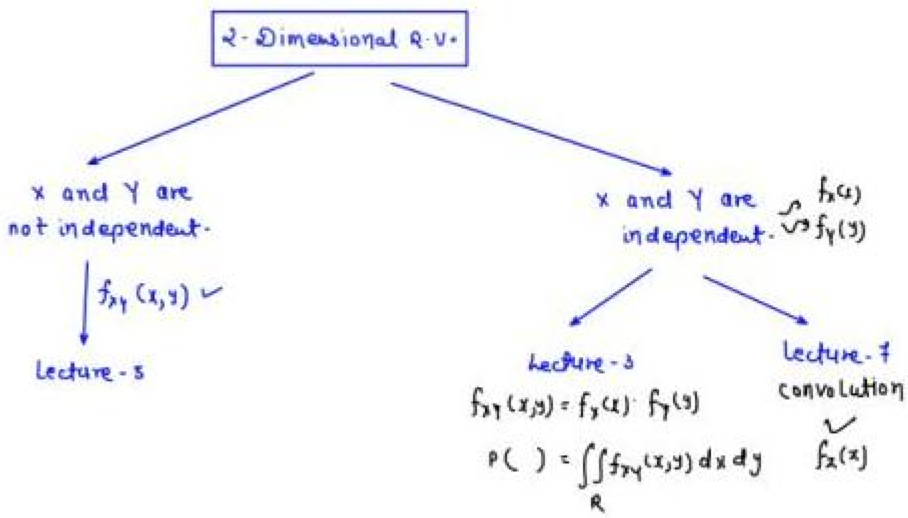
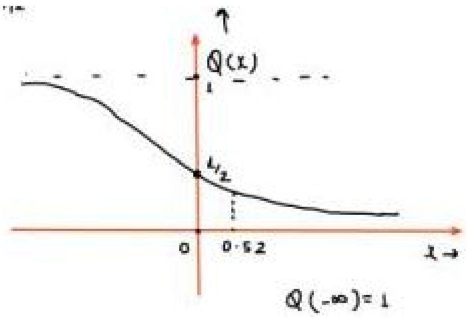


* Central limit theorem:-

If you take a large number of independent and identically distributed (i.i.d.) random variables, their average (or sum) will be approximately normally distributed, no matter what the original distribution was — as long as the variables have a finite mean and variance.

Q. u, v, w, x, y are 5 I.I.O. R.V. ~ u[0,1]

$$R.V.\,Z = U + 2V - W + x - 2Y$$



$$P(Z > 1) = ?$$

(a) 0.5 (b) 0.3 (c) 0.8 (d) 1

$$f_Z(x) \rightarrow Gaussian$$

$$Z \sim N(\mu_z, \sigma_z^2)$$

$$P(U + 2V - W + x - 2Y > 1) = ?$$

$$P(Z > 1) = Q\left(\frac{1 - \mu_z}{\sigma_z}\right)$$

$$Z = U + 2V - W + x - 2Y$$

$$\mu_z = \frac{1}{2} + 2(\frac{1}{2}) - \frac{1}{2} + \frac{1}{2} - 2(\frac{1}{2}) = \frac{1}{2}$$

$$\sigma_z = \sqrt{\frac{1}{2}}$$

$$\sigma_z^2 = \sigma_u^2 + 4\sigma_v^2 + \sigma_w^2 + \sigma_x^2 + 4\sigma_y^2 = \frac{1}{12} + \frac{4}{12} + \frac{1}{12} + \frac{1}{12} + \frac{4}{12} = \frac{1}{12}$$

$$P(Z > 1) = Q\left(\frac{1 - 0.5}{\sqrt{\frac{1}{12}}}\right) = Q(0.52) = ? = 0.3$$

$$Q(0) = 1/2$$

$$Q(0.52) < 1/2$$

$$Q(0.52) < 0.5$$

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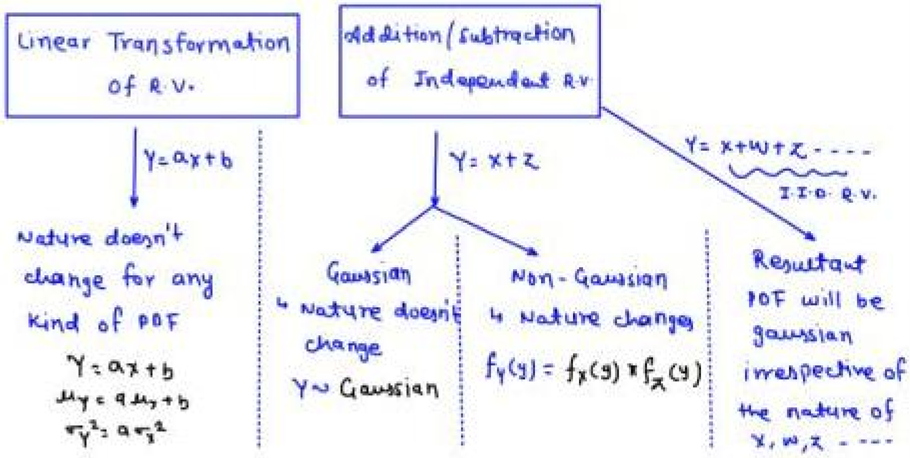
4 Summary:-



4

$$Y = ax + b$$

$$\mu_Y = a\mu_y + b$$

$$\sigma_Y^2 = a^2\sigma_x^2$$

$$Y = x + z$$

$$\mu_Y = \mu_y + \mu_z$$

$$\sigma_Y^2 = \sigma_x^2 + \sigma_z^2 + 2\sigma_{xz}^{cov(x,z)}$$

4 if x, z, w, u are independent R.V.

$$Y = x + z$$

$$\mu_Y = \mu_y + \mu_z$$

$$\sigma_Y^2 = \sigma_x^2 + \sigma_z^2$$

$$V = x + z + w + u$$

$$\mu_V = \mu_y + \mu_z + \mu_w + \mu_v$$

$$\sigma_V^2 = \sigma_x^2 + \sigma_z^2 + \sigma_w^2 + \sigma_v^2$$

4 CRV to CRV Non-Linear Transformation:-

Q

$$f_x(z) = 2e^{-2z}U(z)$$

$$Y = \ln(x + \lambda)$$

$$\text{POF of } Y?$$

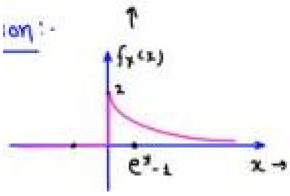
4

$$x \in [0, \infty]$$

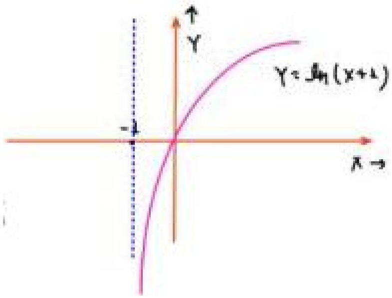
$$Y = \ln(x + \lambda)$$

$$Y_{min} = \ln(\lambda) = 0$$

$$Y_{max} = \ln(\infty) = \infty$$



$$Y \in [0, \infty]$$



COF of Y

$$F_Y(y) = P[Y \leq y] = P[\ln(x+\lambda) \leq y]$$

$$= P[x \leq e^{y-\lambda}]$$

$$= \int_{-\infty}^{e^{y-\lambda}} f_x(z)\, dz$$

$$F_Y(y) = 1 - e^{-2(e^{y-\lambda})}; 0 < y < \infty$$

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

$$= \int_0^{e^{y-\lambda}} ae^{-2x}\, dx = \frac{2}{\lambda}[e^{-2x}]e^{y-\lambda}$$

$$= -e^{-2(e^{y-\lambda})} \frac{d}{dy}[-2(e^{y-\lambda})]$$

$$f_Y(y) = ae^y e^{-2(e^{y-\lambda})U(y)}$$

$$= [1 - e^{-2(e^{y-\lambda})}]$$

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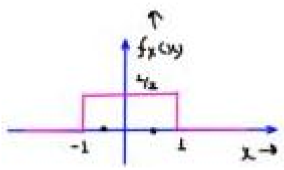
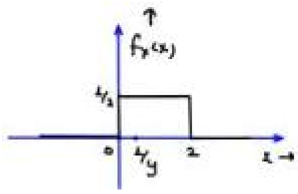




Q.

$$x \sim \cup[0,2]$$

$$\gamma = \frac{1}{x}$$

$$\text{PDF of } \gamma?$$

$$x \in [0,2]$$

$$y \in [\frac{1}{2}, \infty]$$

$$F_Y(y) = P[Y \le y] = P[\frac{1}{x} \le y] = P[x \ge \frac{1}{y}] = \int_{\frac{1}{y}}^{2} f_X(x)\, dx = \frac{1}{2}[2 - \frac{1}{y}]$$

$$F_Y(y) = 1 - \frac{1}{2y} \quad ; \quad \frac{1}{2} < y < \infty$$

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{1}{2y^2} \quad ; \quad \frac{1}{2} < y < \infty$$

Q.

$$x \sim \cup[-1,1]$$

$$\gamma = x^2$$

$$\text{PDF of } \gamma = ?$$

$$y \in [0,1]$$

$$F_Y(y) = P[Y \le y] = P[x^2 \le y] = P[-\sqrt{y} < x < \sqrt{y}] = \int_{\sqrt{y}}^{\sqrt{y}} f_X(x)\, dx = \frac{1}{2}[2\sqrt{y}] = \sqrt{y}$$

$$F_Y(y) = \sqrt{y} \quad ; \quad 0 < y < 1$$

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{1}{2\sqrt{y}} \quad ; \quad 0 < y < 1$$

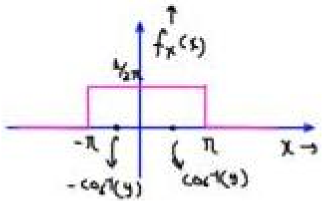
Q.

$$x \sim \cup[0,\pi]$$

$$\gamma = \cos x$$

$$\text{PDF of } \gamma?$$

$$y \in [-1,1]$$



$$F_Y(y) = P[Y \le y] = P[\cos x \le y]$$

$$F_Y(y) = 1 - \frac{1}{\pi}\cos^{-1} y$$

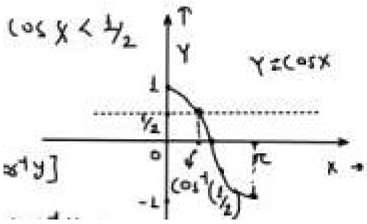
$$-1 < y < 1$$

$$= P[x \ge \cos^{-1}(y)]$$

$$= \int_{\pi}^{x} f_X(x)\,dx = \frac{1}{\pi}[\pi - \cos^{-1} y]$$

$$f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}} \quad ; \quad -1 < y < 1$$

$$= 1 - 1/\pi \cos^{-1} y \quad ; \quad -1 < y < 1$$



Q.

$$x \sim \cup[-\pi, \pi]$$

$$\gamma = \cos x$$

$$\text{PDF of } \gamma = ?$$

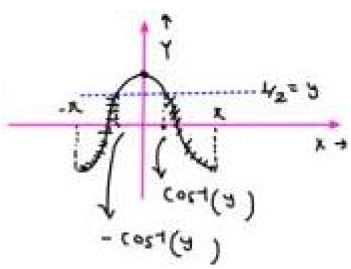
$$y \in [-1, 1]$$

$$F_Y(y) = P[Y \le y] = P[\cos x \le y]$$

$$= P[x \ge \cos^{-1}(y)] + P[x \le -\cos^{-1}(y)]$$

$$= \frac{\pi - \cos^{-1} y}{2\pi} + \frac{\pi - \cos^{-1} y}{2\pi}$$

$$\boxed{\begin{array}{l} F_Y(y) = 1 - \frac{\cos^{-1}(y)}{\pi} \quad ; \quad -1 < y < 1 \\ f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}} \quad ; \quad -1 < y < 1 \end{array}}$$



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U and V are I.E.O. R.V. with zero mean.

COF of U and 2V are F(x) and G(x) respectively.

Correct statement of all values of x,

$F(x) - G(x) \leq 0$

$F(x) - G(x) \geq 0$

(c) $[F(x) - G(x)] \cdot x \leq 0$

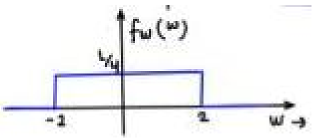
$[F(x) - G(x)] \cdot x > 0$

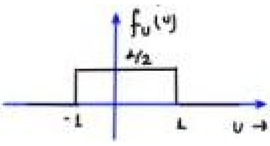
4

U ~ Uniform [-1, 1]

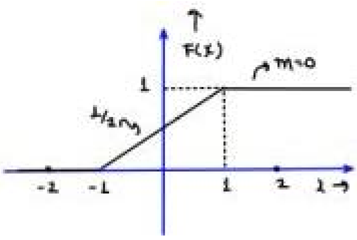
V ~ Uniform [-1, 1]

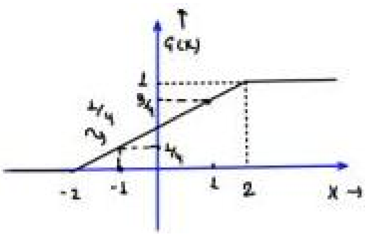
W = 2V

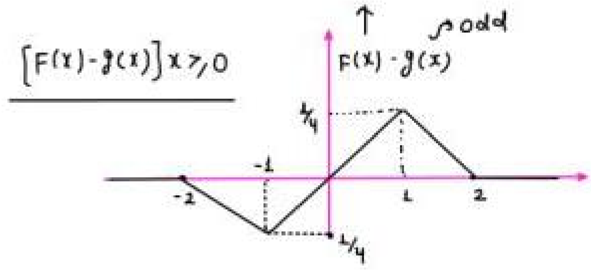


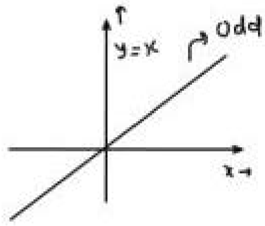


COF = ∫ P OF









$x_1 + x_2 + x_3 \in [0, 3]$

Q. Let x_1, x_2, x_3 i.i.d. R.V. ~ U[0, 1]

GATE 2014

(a) $P[x_1 + x_2 + x_3] = 1/6$
(b) $P[x_1 + x_2 + x_3 \leq 1] = 1/6$
(c) $P[x_1 + x_2 + x_3 \leq 2] = 5/6$
(d) $P[x_1 + x_2 + x_3 \leq 3] = 1$
(e) $P[x_1 + x_2 + x_3 \leq 0] = 0$

$\mu = 1/2$

$\sigma^2 = 1/12$

$x = x_1$

$y = x_2$

$z = x_3$ $Q(0) = 1/2$; $Q(1) = 0.16$

$Q(\infty) = 0$; $Q(1) = 0.84$

$Q(-\infty) = 1$

(a) $P[x + y < z] = P[x + y - z < 0] = P[\omega < 0] = 1 - P(\omega > 0)$

Gauasian

$\mu\_w = \mu\_x + \mu\_y - \mu\_z = 1/2$

$\sigma\_w^2 = \sigma\_x^2 + \sigma\_y^2 + \sigma\_z^2 = 3x\ 1/12 = 1/4$

$= 1 - Q(0 - \mu\_z / \sigma\_x) = 1 - Q(-1)$

= Q(1)

σ_w = 1/2

= 0.16

4

f_x(x) = { 1 ; 0 < x < 1
0 ; 0/ω

f_y(y) = { 1 ; 0 < y < 1
0 ; 0/ω

x, y, z → independent

f_{xyz}(x, y, z) = f_y(x) · f_y(y) · f_z(x)

f_z(x) = { 1 ; 0 < z < 1
0 ; 0/ω

f_{yz}(x, y, z) = { 1 ; 0 < x < 1 , 0 < y < 1 , 0 < z < 1
0 ; 0/ω

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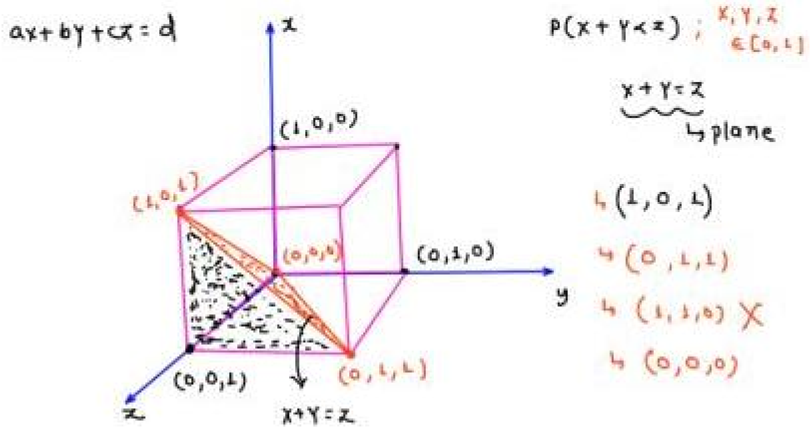
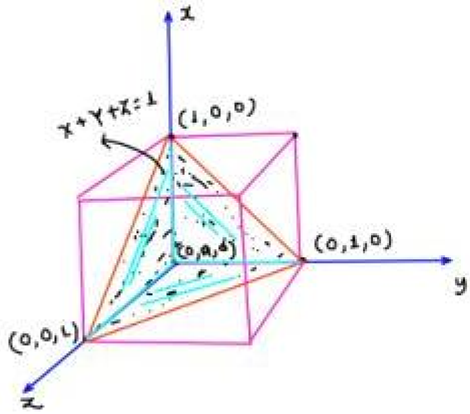


$$V = \frac{1}{3}(\text{drea of base}) \times \text{Height} = \frac{1}{3} \times [\frac{1}{2} \times 1 \times 1] \times 1 = \frac{1}{6}$$

$$P(x+y<z) = \iiint_V f_{xyz}(x,y,z)\,dx\,dy\,dz$$
$$= \iiint_V dx\,dy\,dz = \frac{1}{6}$$

$$P(x+y<z) = \frac{1}{6} = 0.167$$

$$\therefore P(x+y+z \leq 1)$$



$x+y+z \leq 1$

$x+y+z = 1$

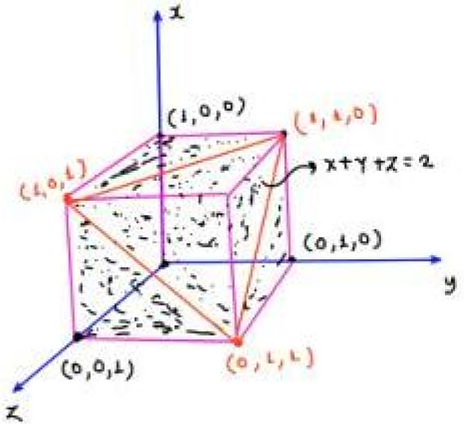
↳ (1,0,0)

↳ (0,1,0)

↳ (0,0,1)

$0+0+0 \leq 1$ ✓

$$P(x+y+z<1) = \iiint_V f_{xyz}(x,y,z)\,dx\,dy\,dz = \iiint_V dx\,dy\,dz$$
$$= \frac{1}{3}(\frac{1}{2})(1) = \frac{1}{6}$$



$x+y+z \leq 2$

$x+y+z = 2$

(1,0,1)

(0,1,1)

(1,1,0)

$0+0+0 \leq 2$

$0 \leq 2$ ✓

$$\iiint_V f_{xyz}(x,y,z)\,dx\,dy\,dz = 1 - \left(\frac{1}{3} \times \frac{1}{2} \times 1\right) = \boxed{\frac{5}{6}} \text{ans}$$

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N.B. - Poisson and binomial distribution will be studied before information theory / Engineering mathematics.